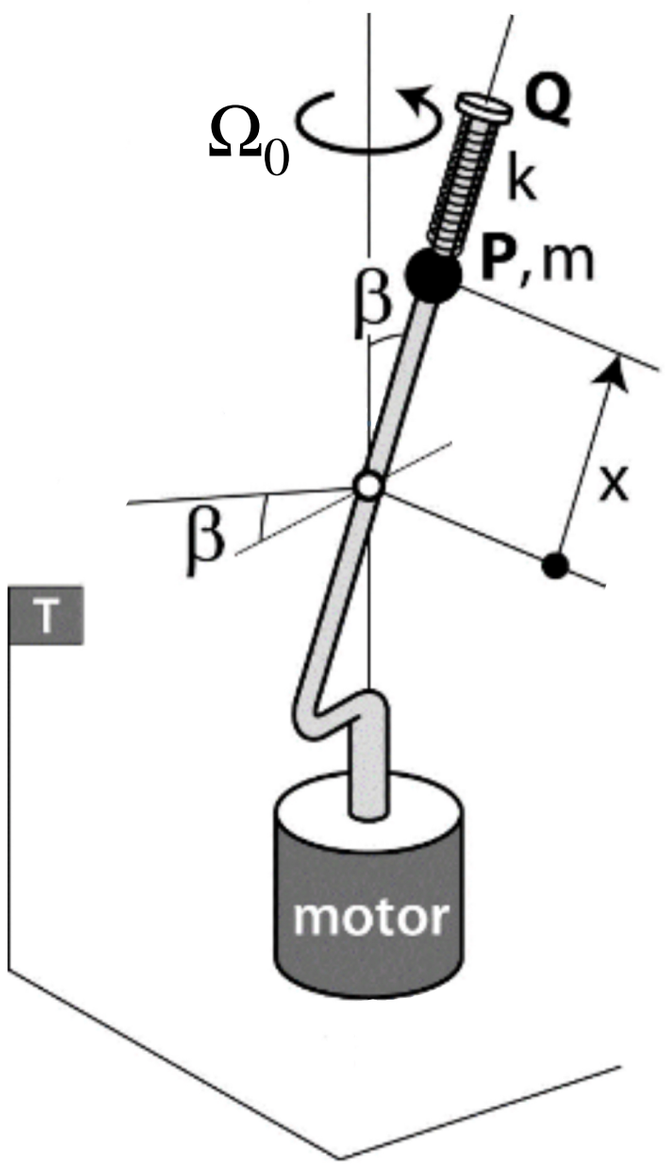
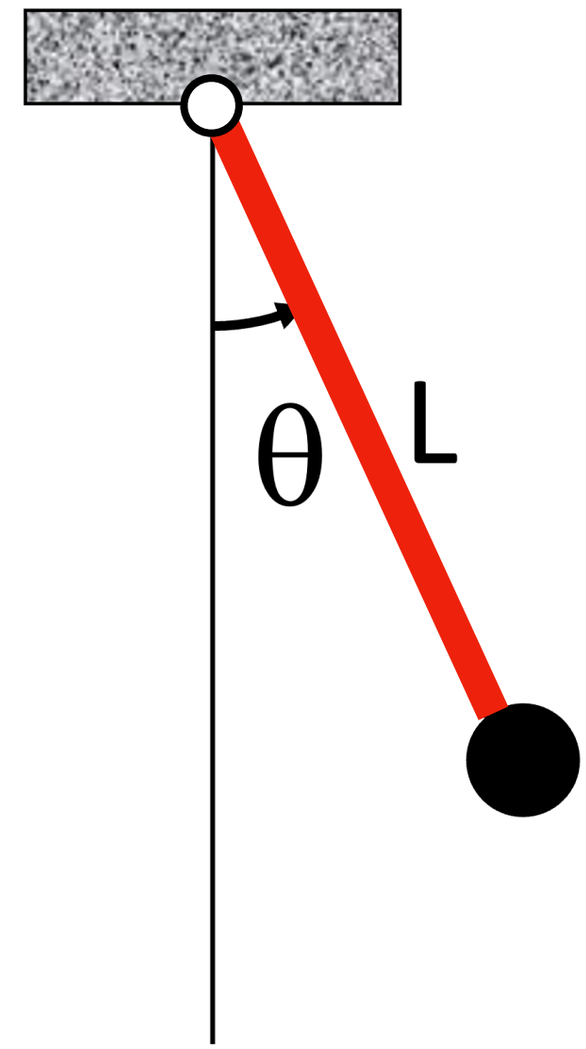
9P

Geometria de masses

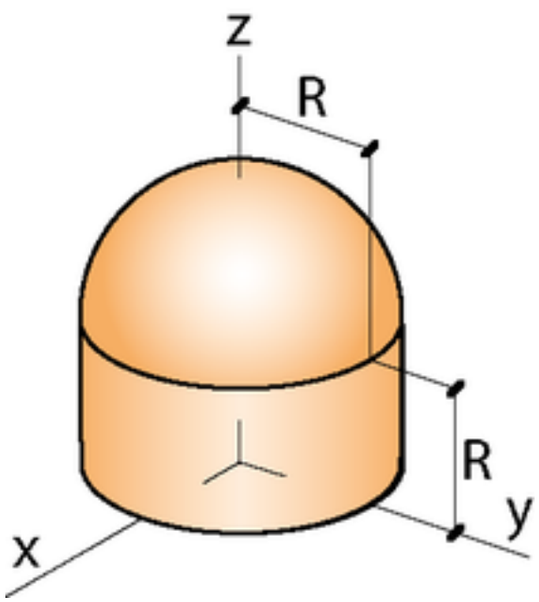
Massa-molla-guia

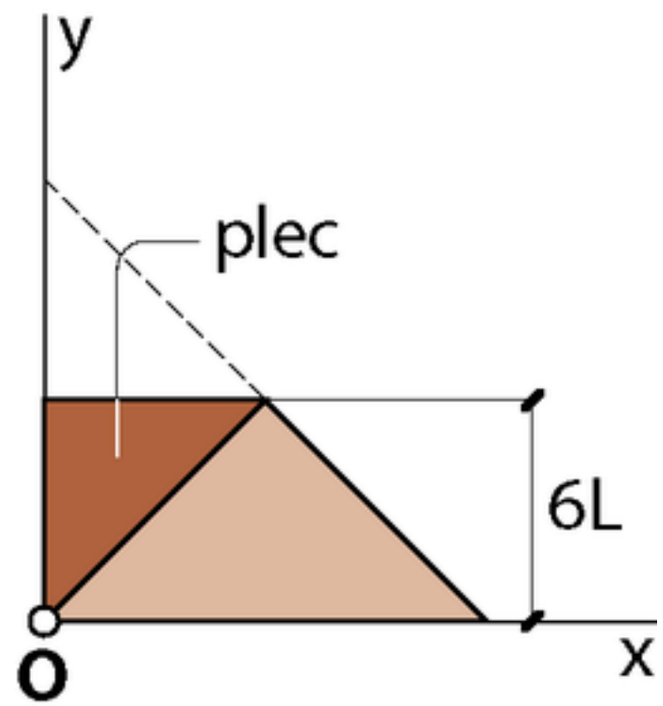


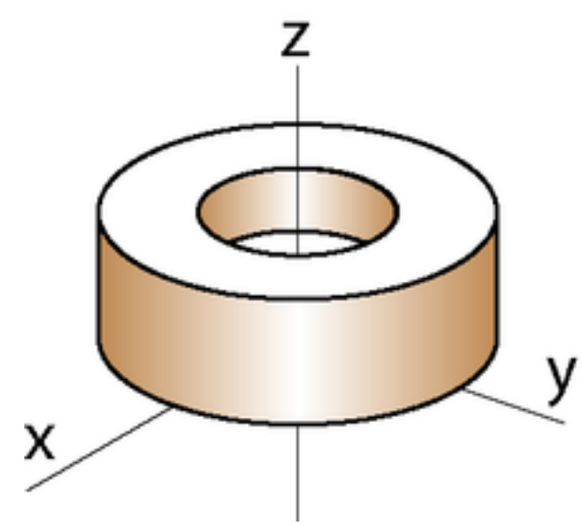
Pèndol simple



Exm: D5.1 - D5.2 - D5.3 (Wikimec)







Tensor d'inèrcia de S a **Q**

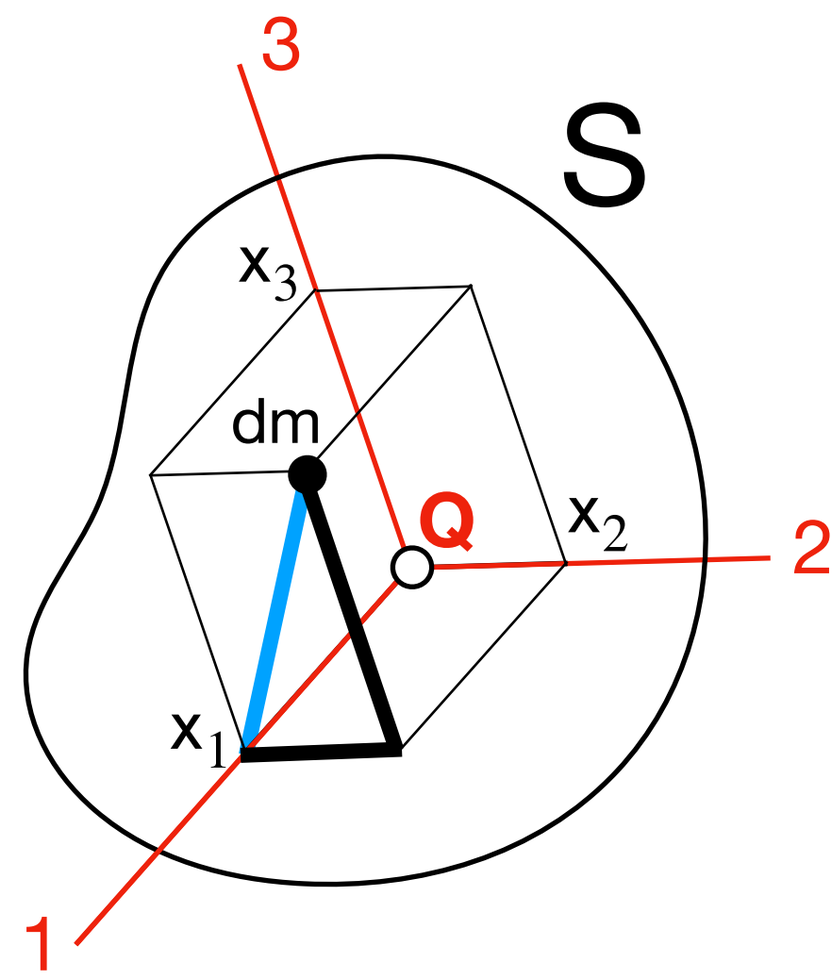
$$[\mathbb{I}(\mathbf{Q})]_B = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix}$$

Moments d'inèrcia

$$I_{ii} = \int_S \underbrace{\left(x_j^2 + x_k^2\right)}_{(\text{ dist a } \textbf{eix i})^2} \, dm \quad \boxed{\geq 0}$$

Exm:

$$I_{11} = \int_S \underbrace{\left(x_2^2 + x_3^2\right)}_{(\text{ dist a } \textbf{eix 1})^2} \, dm$$



B = (1,2,3)

Tensor d'inèrcia de S a **Q**

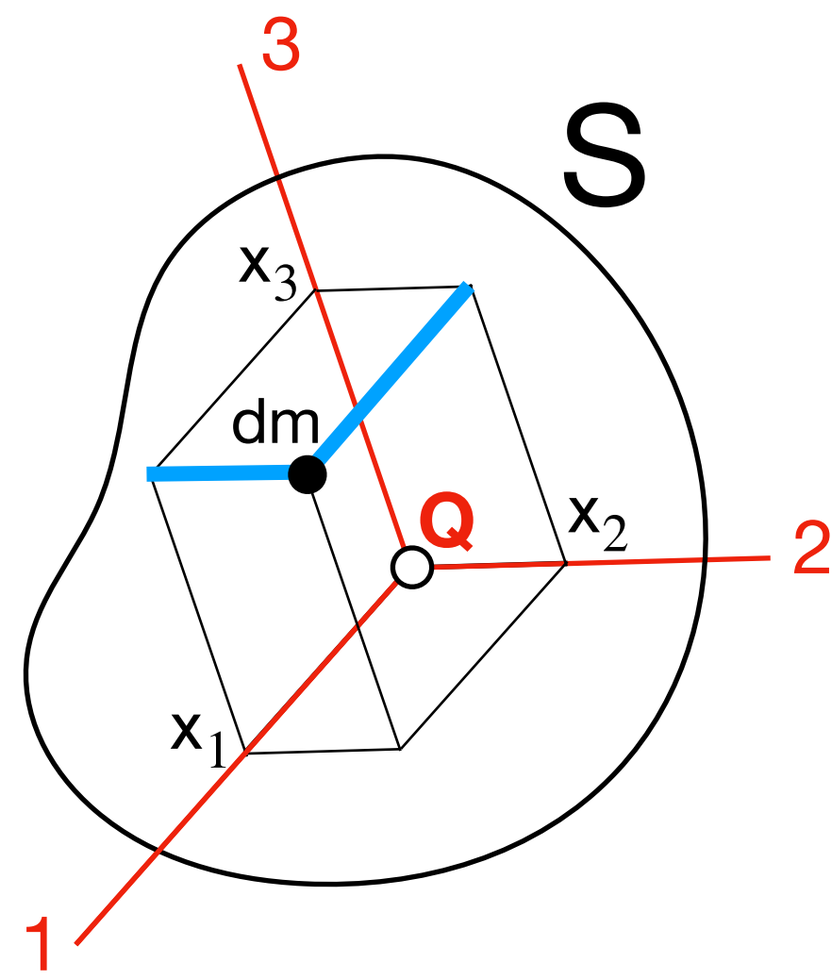
$$[\mathbb{I}(\mathbf{Q})]_B = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix}$$

Productes d'inèrcia

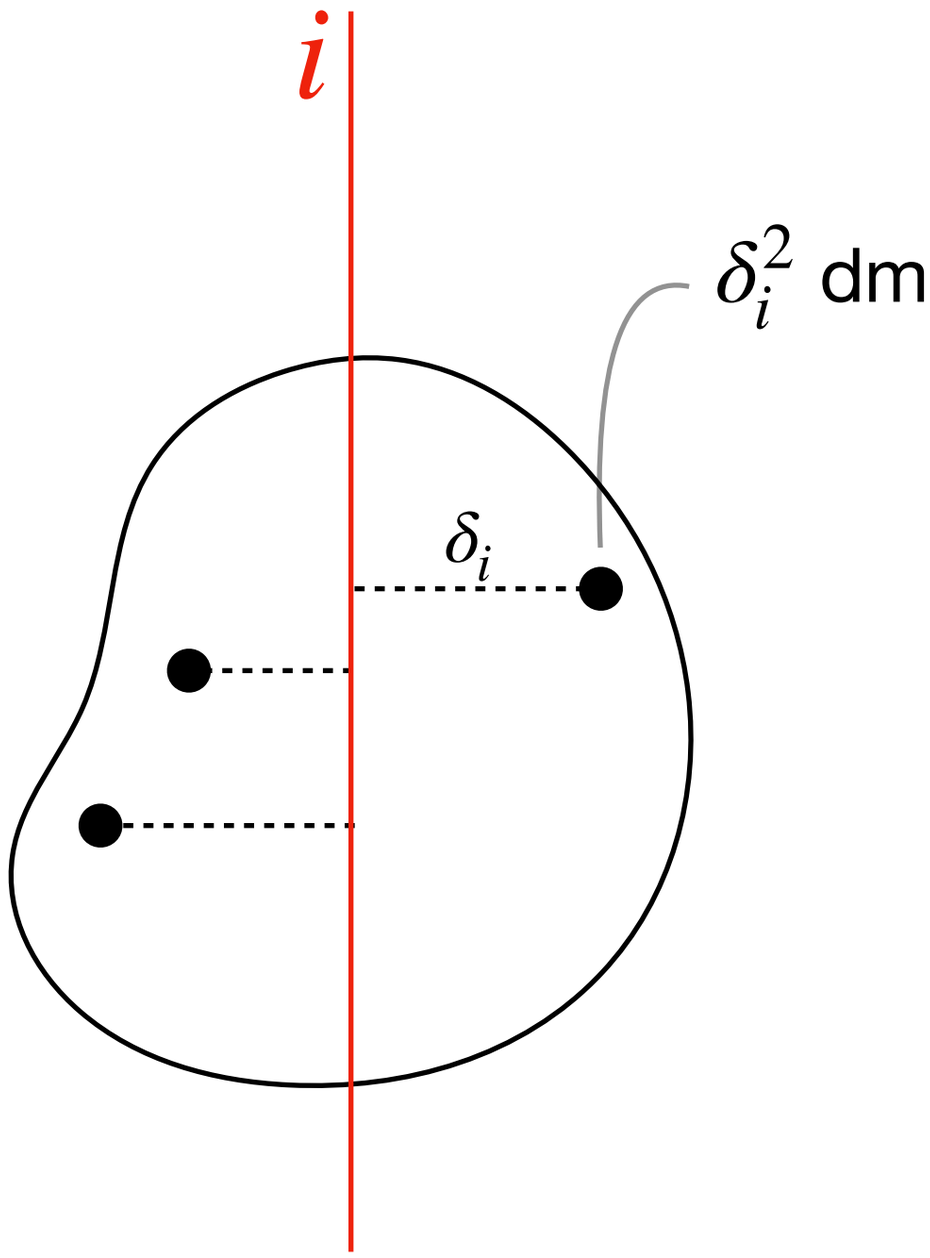
$$I_{ij} = -\int_S x_i \, x_j \, dm$$ (>0, <0, =0)

Exm:

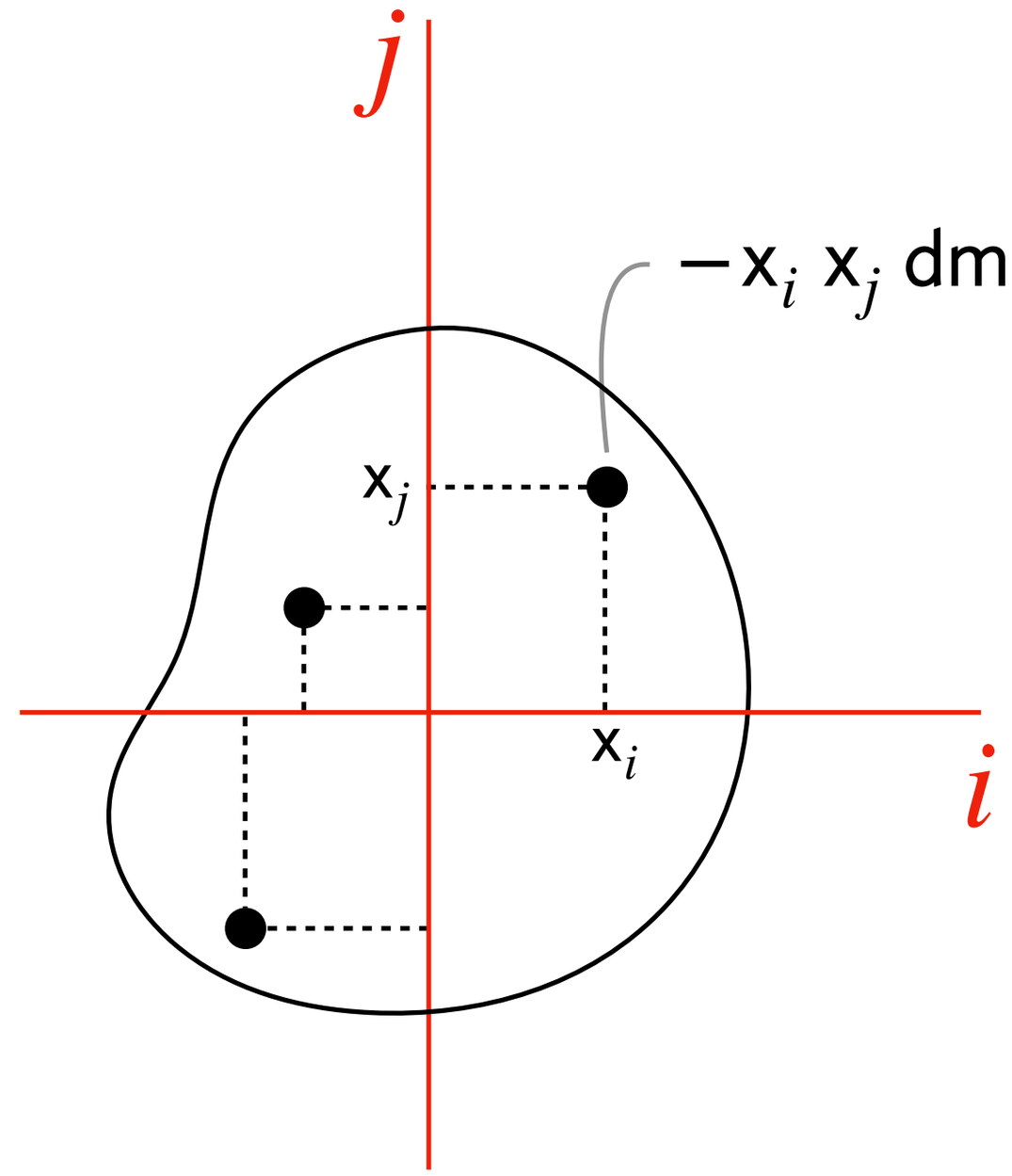
$$I_{12} = -\int_S x_1 \, x_2 \, dm$$



Mom. inèrcia I_{ii}



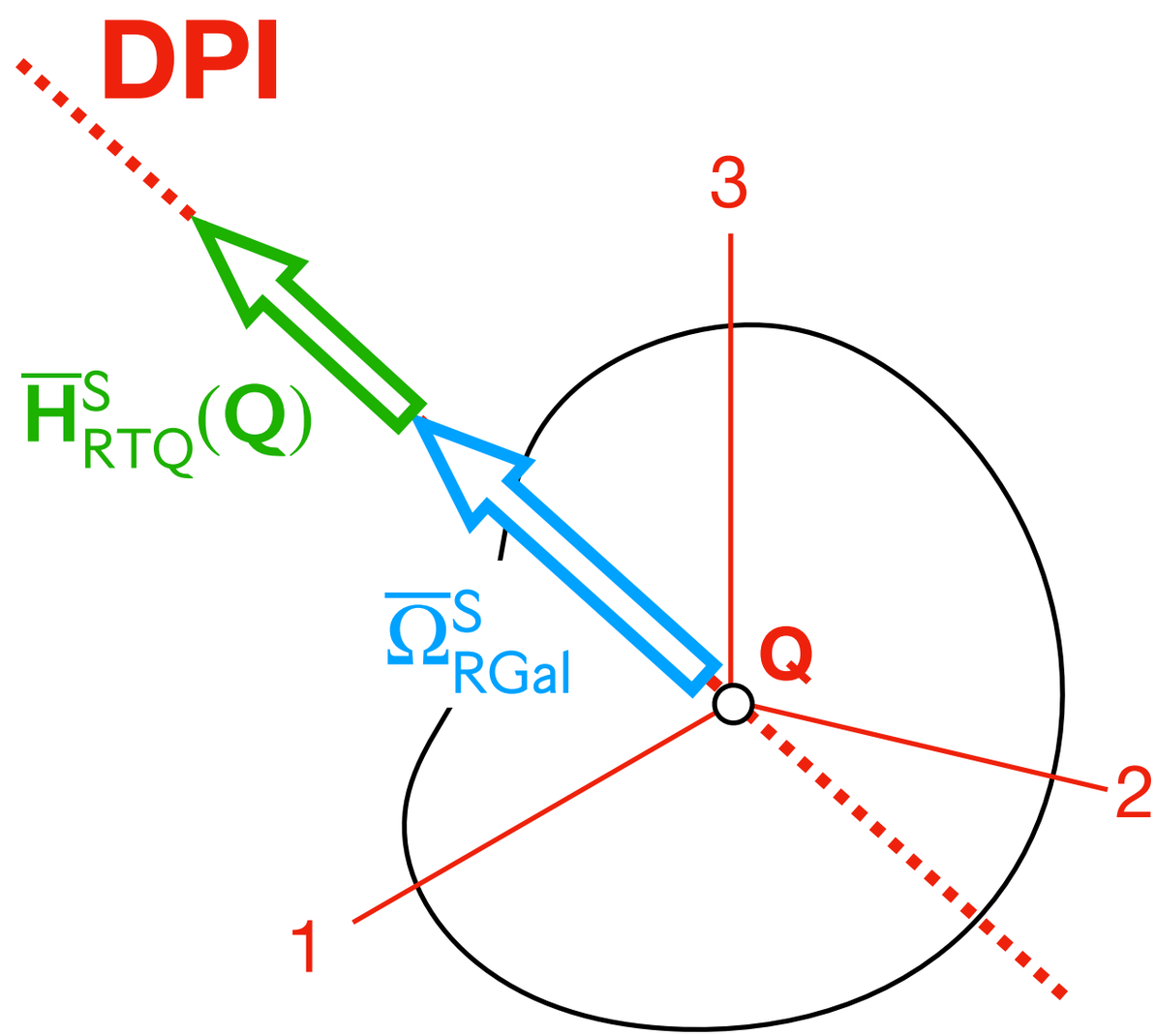
Prod. inèrcia I_{ij}



Direccions i moments principals d'inèrcia

$$\overline{\mathbf{H}}^{S}_{RTQ}(\mathbf{Q}) = \mathbb{I}(\mathbf{Q})\ \overline{\Omega}^{S}_{RGal}$$

Quan són || diem que la seva dir. és **DPI**



$$\overline{\mathbf{H}}^{S}_{RTQ}(\mathbf{Q}) = \lambda\ \overline{\Omega}^{S}_{RGal}$$

MPI associat a la DPI

Les direccions de la base ... quan són **DPI** I?

Perquè la **dir 1** sigui DPI cal que:

$$\begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix} \begin{Bmatrix} \Omega \\ 0 \\ 0 \end{Bmatrix} = \lambda \begin{Bmatrix} \Omega \\ 0 \\ 0 \end{Bmatrix}$$

Ergo:

$$I_{11}\,\Omega = \lambda\,\Omega \Rightarrow \lambda = I_{11}$$

$$I_{12}\,\Omega = 0 \Rightarrow I_{12} = 0$$

$$I_{13}\,\Omega = 0 \Rightarrow I_{13} = 0$$

$\Longrightarrow$

Ha de ser

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & I_{22} & I_{23} \\ 0 & I_{23} & I_{33} \end{bmatrix}$$

Les direccions de la base ... quan són **DPI** I?

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \mathtt{I}_{22} & \mathtt{I}_{23} \\ 0 & \mathtt{I}_{23} & \mathtt{I}_{33} \end{bmatrix} \iff$$ **Dir 1** és DPI

$$\begin{bmatrix} \mathtt{I}_{11} & 0 & \mathtt{I}_{13} \\ 0 & \lambda & 0 \\ \mathtt{I}_{13} & 0 & \mathtt{I}_{33} \end{bmatrix} \iff$$ **Dir 2** és DPI

$$\begin{bmatrix} \mathtt{I}_{11} & \mathtt{I}_{12} & 0 \\ \mathtt{I}_{12} & \mathtt{I}_{22} & 0 \\ 0 & 0 & \lambda \end{bmatrix} \iff$$ **Dir 3** és DPI

Bases adients?

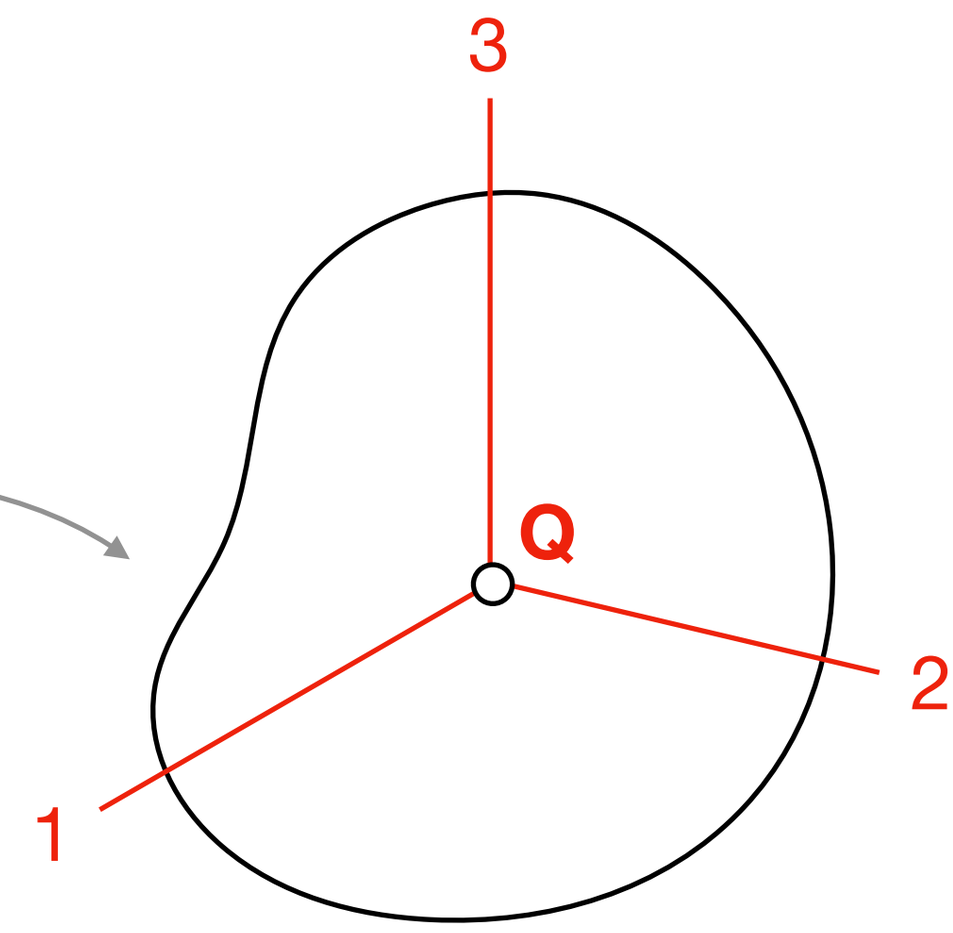
$$[\mathbb{I}(\mathbf{Q})]_{\mathrm{B}} = \begin{bmatrix} \mathrm{I}_{11} & \mathrm{I}_{12} & \mathrm{I}_{13} \\ \mathrm{I}_{12} & \mathrm{I}_{22} & \mathrm{I}_{23} \\ \mathrm{I}_{13} & \mathrm{I}_{23} & \mathrm{I}_{33} \end{bmatrix}$$

Utilitzarem bases B

en les quals $\mathbb{I}(\mathbf{Q})$ és **constant**

Per exemple:

B fixa al sòlid

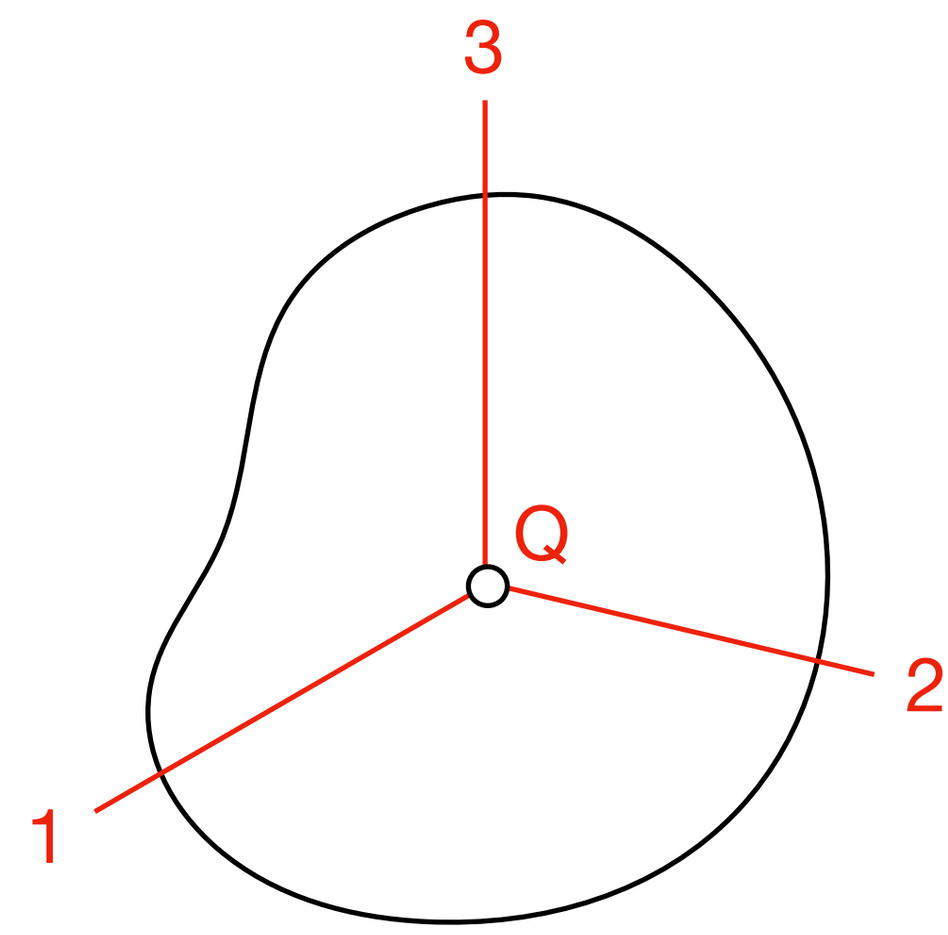


$$[\mathbb{I}(\mathbf{Q})]_{\mathrm{B}} = \begin{bmatrix} \mathtt{I}_{11} & \mathtt{I}_{12} & \mathtt{I}_{13} \\ \mathtt{I}_{12} & \mathtt{I}_{22} & \mathtt{I}_{23} \\ \mathtt{I}_{13} & \mathtt{I}_{23} & \mathtt{I}_{33} \end{bmatrix}$$

Objectiu d'avui:

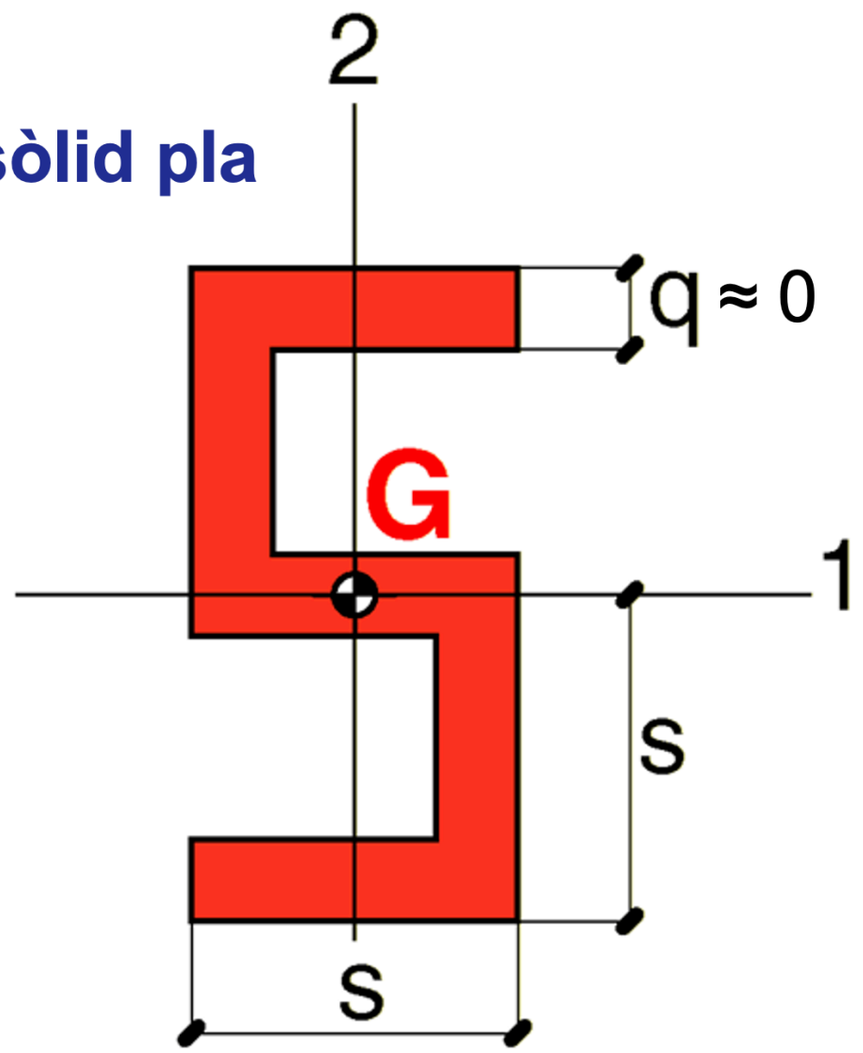
Aprendre a **avaluar-lo**

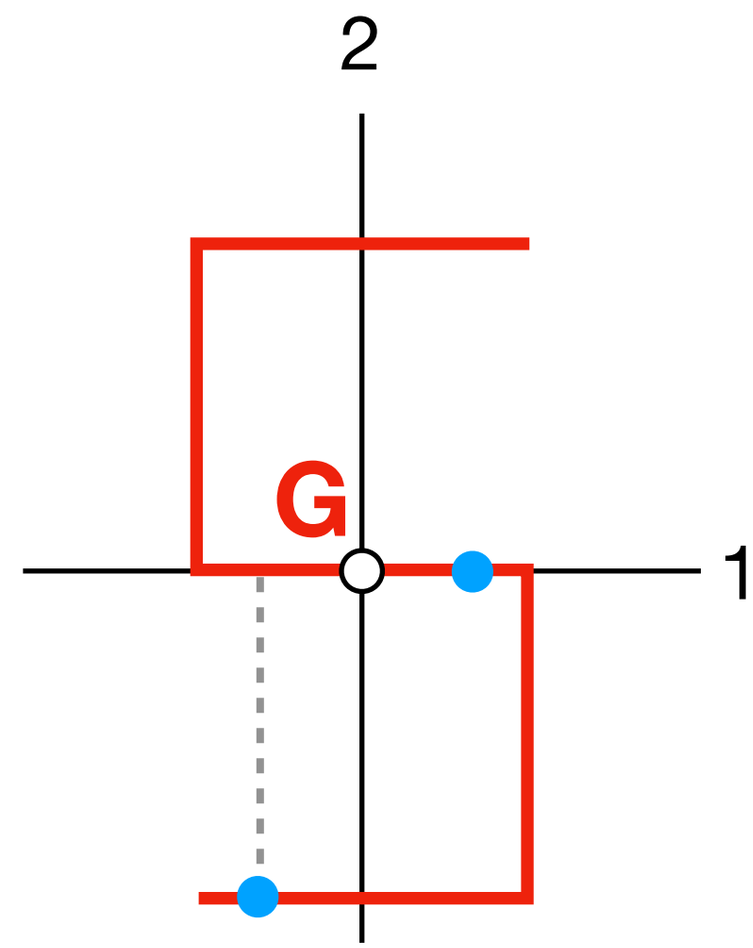
- Qualitativament
- Quantitativament



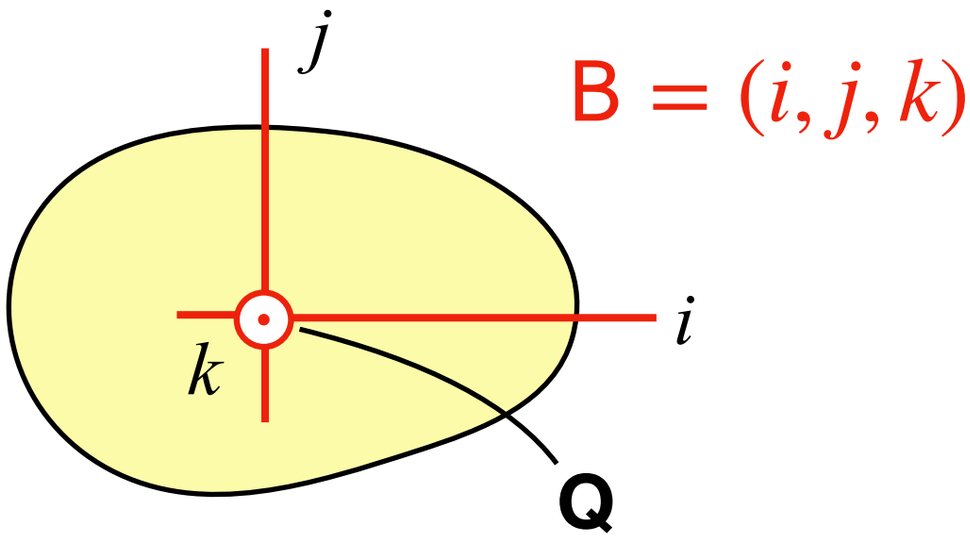
$$\left[\mathbb{I}(\mathbf{G})\right]_B \text{ qualitatiu ?}$$

sòlid pla





Per un sòlid pla



$\mathrm{B} = (i, j, k)$

es compleix

1

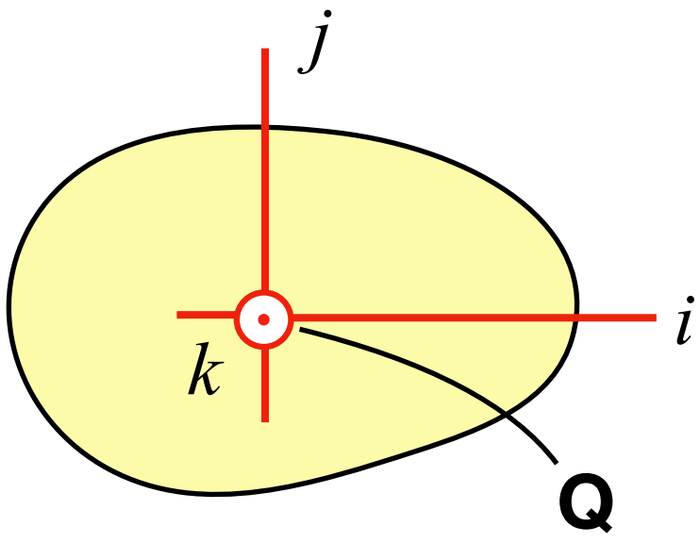
$\forall \mathbf{Q} \in$ sòlid, la **dir. $\perp$ al sòlid** és **DPI**

$$\left[\mathbb{I}(\mathbf{Q})\right]_{\mathrm{B}} = \begin{bmatrix} \mathrm{I}_{ii} & \mathrm{I}_{ij} & 0 \\ \mathrm{I}_{ij} & \mathrm{I}_{jj} & 0 \\ 0 & 0 & \mathrm{I}_{kk} \end{bmatrix}$$

$\mathrm{I}_{kk} = \mathrm{I}_{ii} + \mathrm{I}_{jj}$

I_{kk} és el **MPI** d'aquesta **DPI**

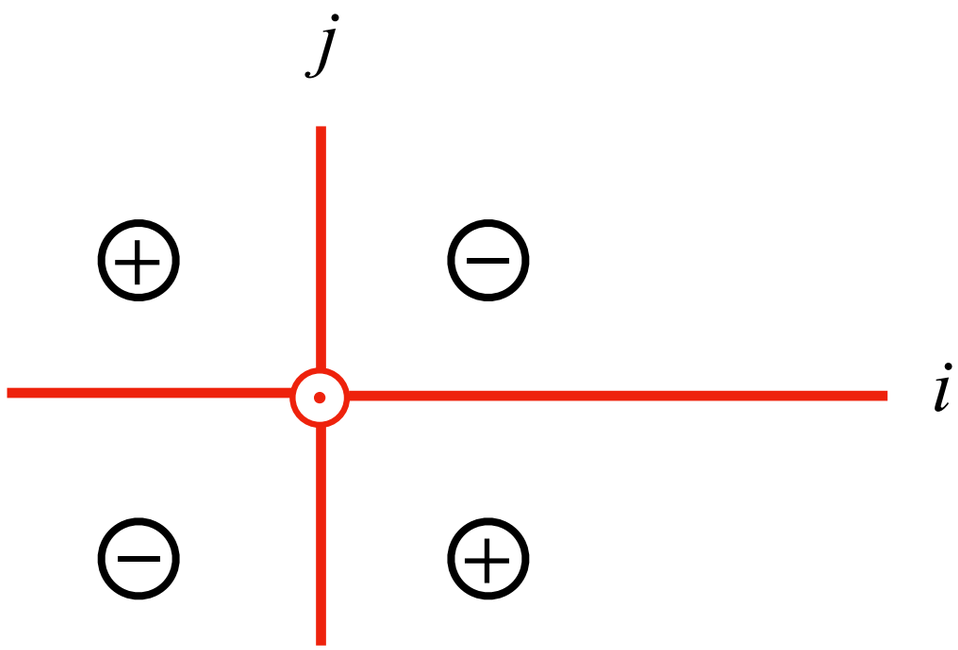
Per un sòlid pla



es compleix

2

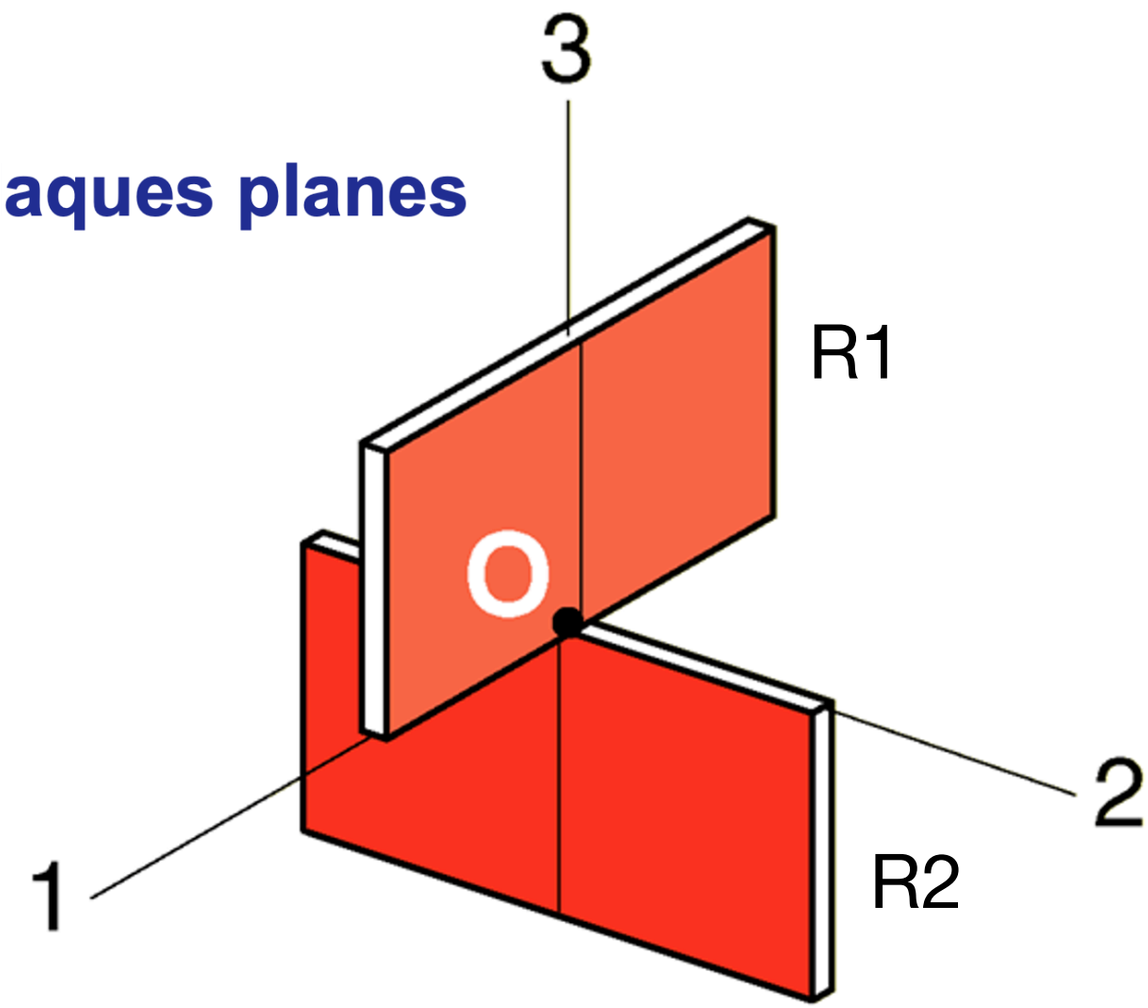
Signe de la contribució dels dm a $\mathtt{I}_{ij}$



[II(O)] ?

qualitatiu

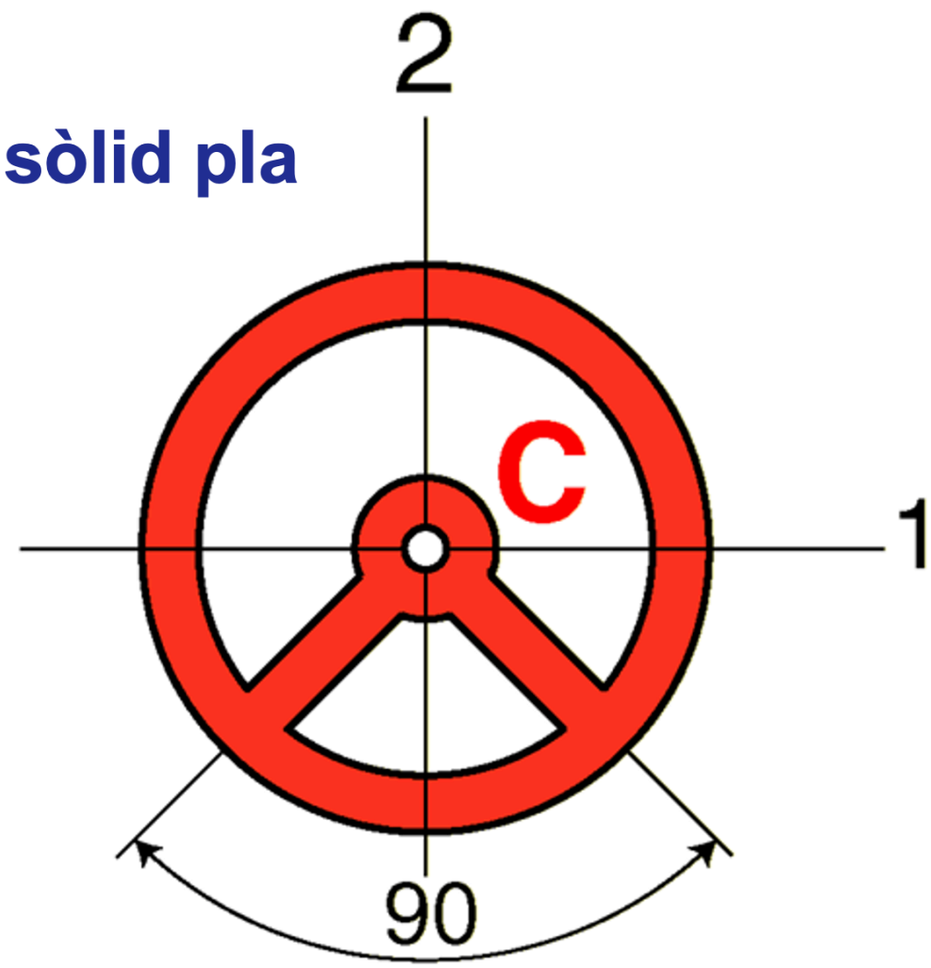
plaques planes



[II(C)] ?

qualitatiu

sòlid pla

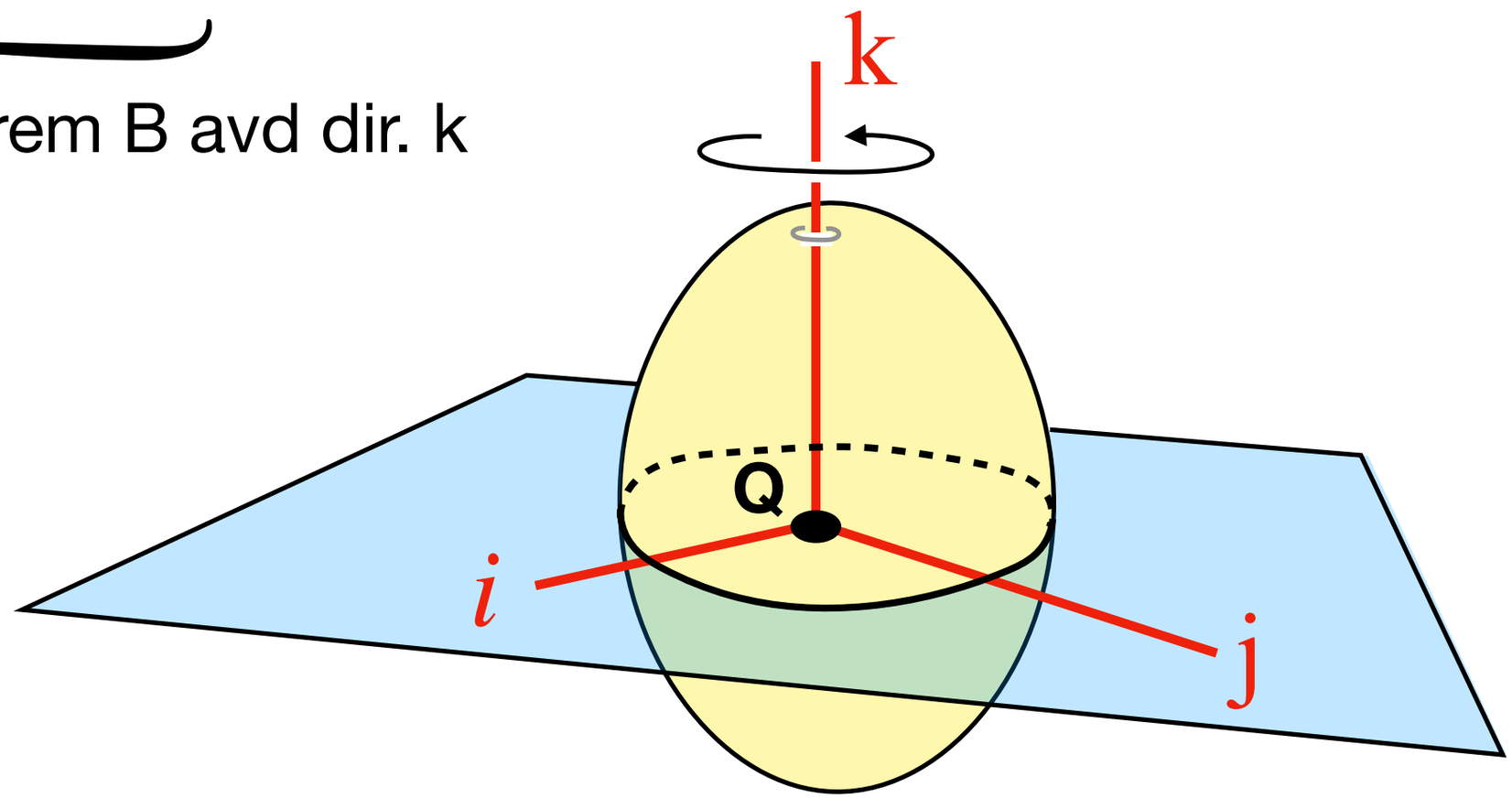


"Rotor simètric per **Q**" en el pla (i, j)

Si per al punt **Q** les dirs. (i, j)
són DPI amb mateix MPI ...

$$[\mathbb{I}(\mathbf{Q})]_B = \begin{bmatrix} \mathrm{I} & 0 & 0 \\ 0 & \mathrm{I} & 0 \\ 0 & 0 & \mathrm{I}_{kk} \end{bmatrix}$$

no canvia si girem B avd dir. k



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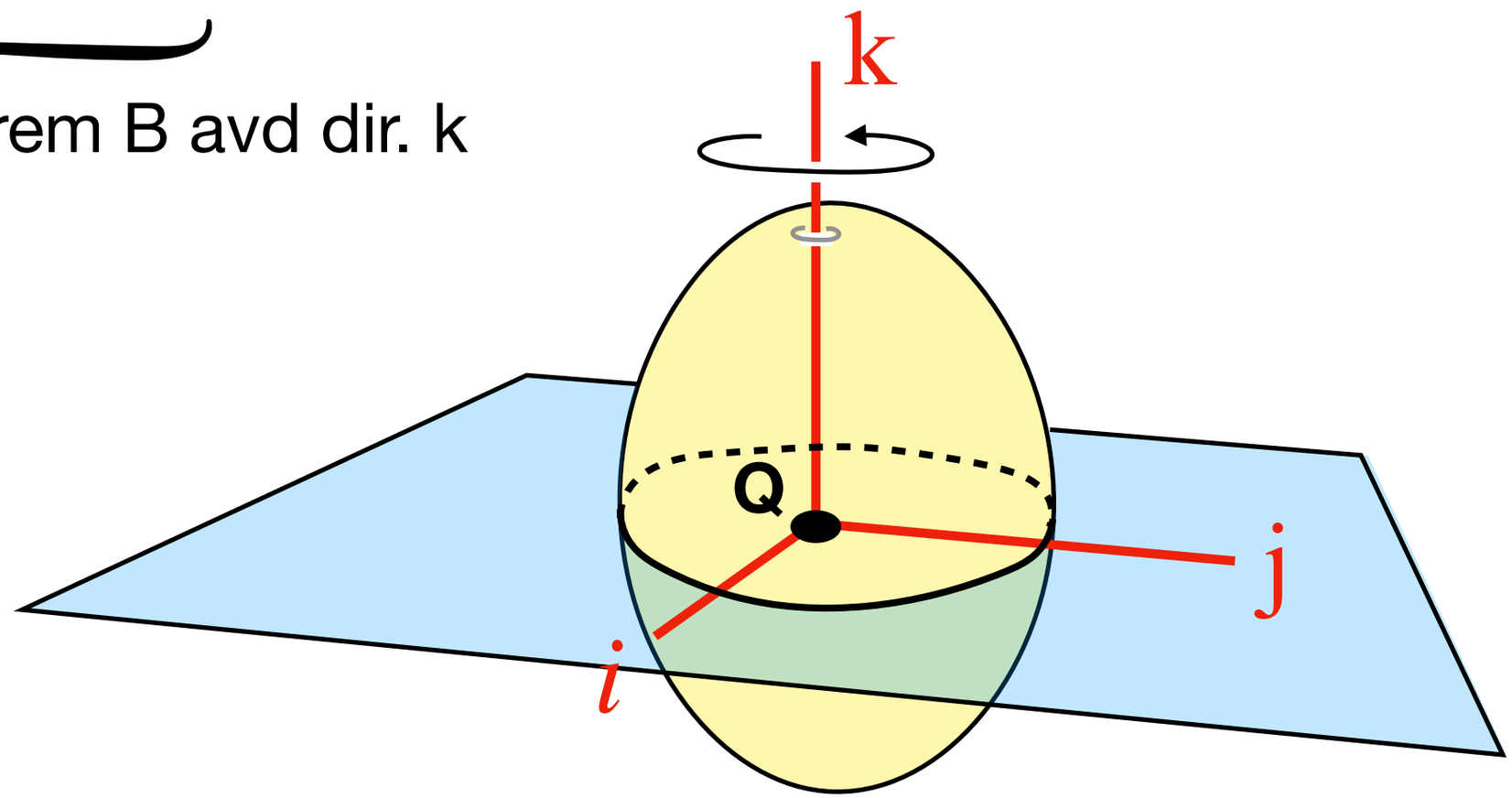
no canvia si girem B avd dir. k



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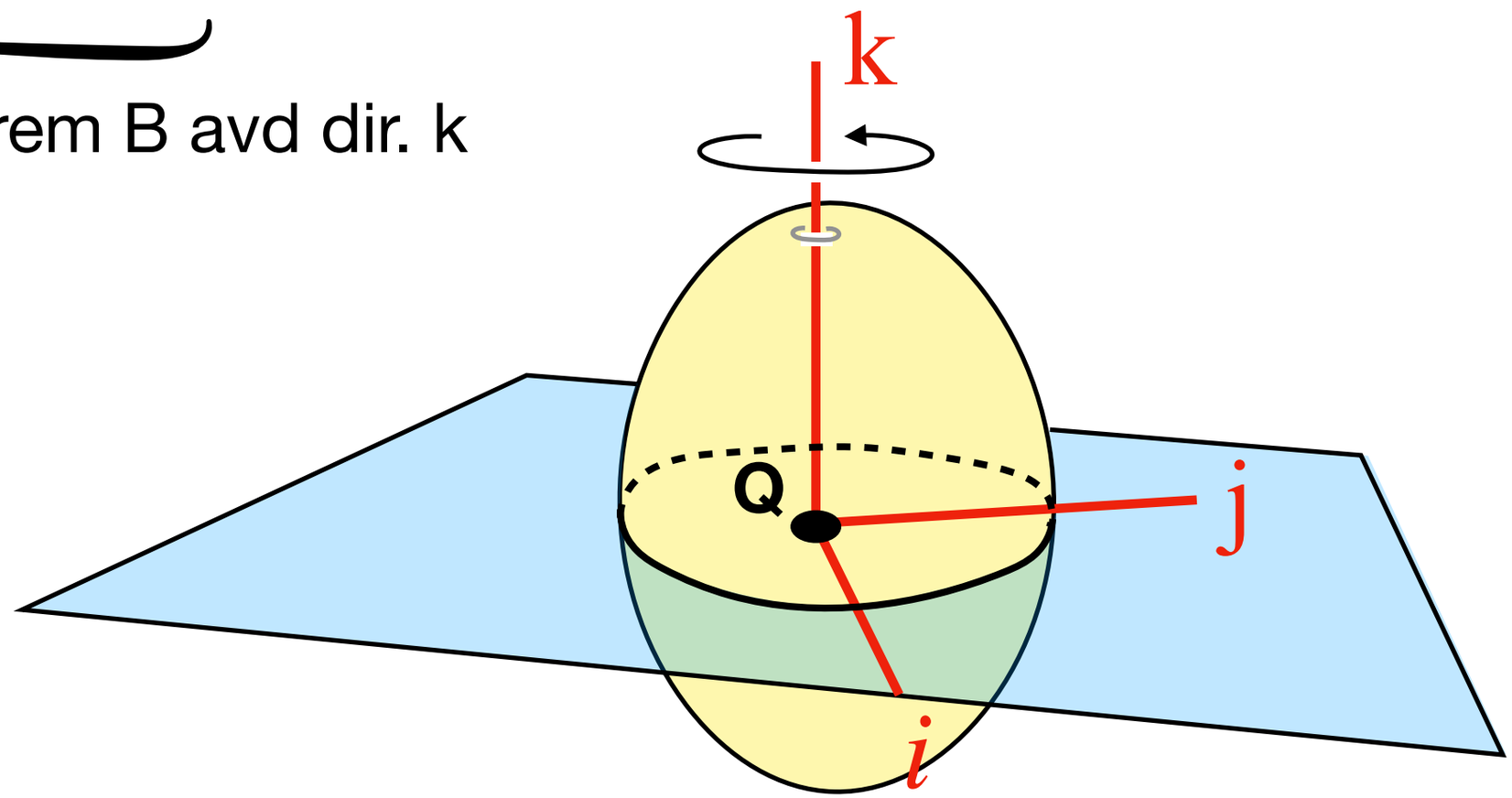
no canvia si girem B avd dir. k

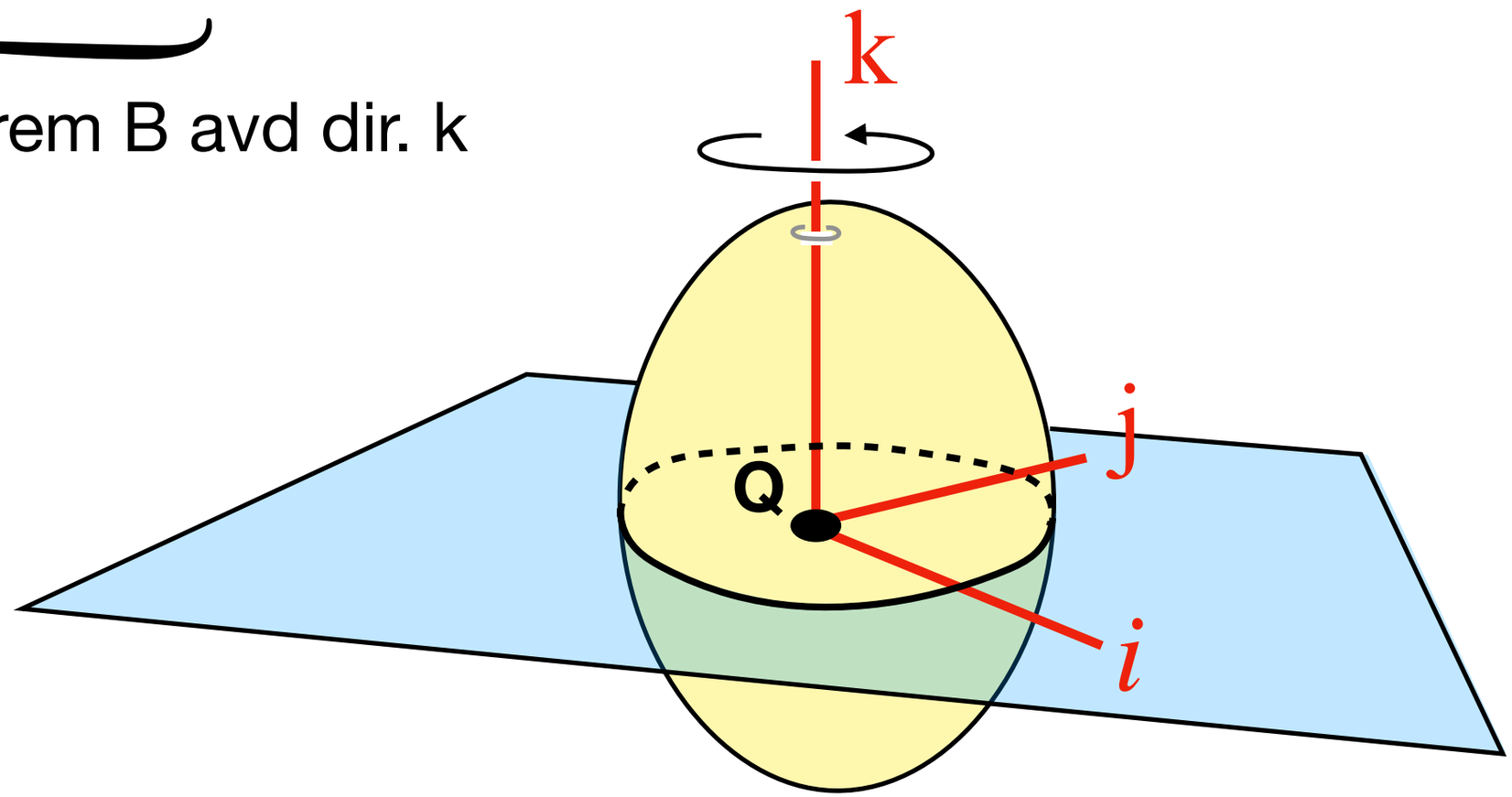


"Rotor simètric per **Q**" en el pla (i, j)

Si per al punt **Q** les dirs. (i, j)
són DPI amb mateix MPI ...

$$[\mathbb{I}(\mathbf{Q})]_{\mathrm{B}} = \begin{bmatrix} \mathrm{I} & 0 & 0 \\ 0 & \mathrm{I} & 0 \\ 0 & 0 & \mathrm{I}_{kk} \end{bmatrix}$$

no canvia si girem B avd dir. k



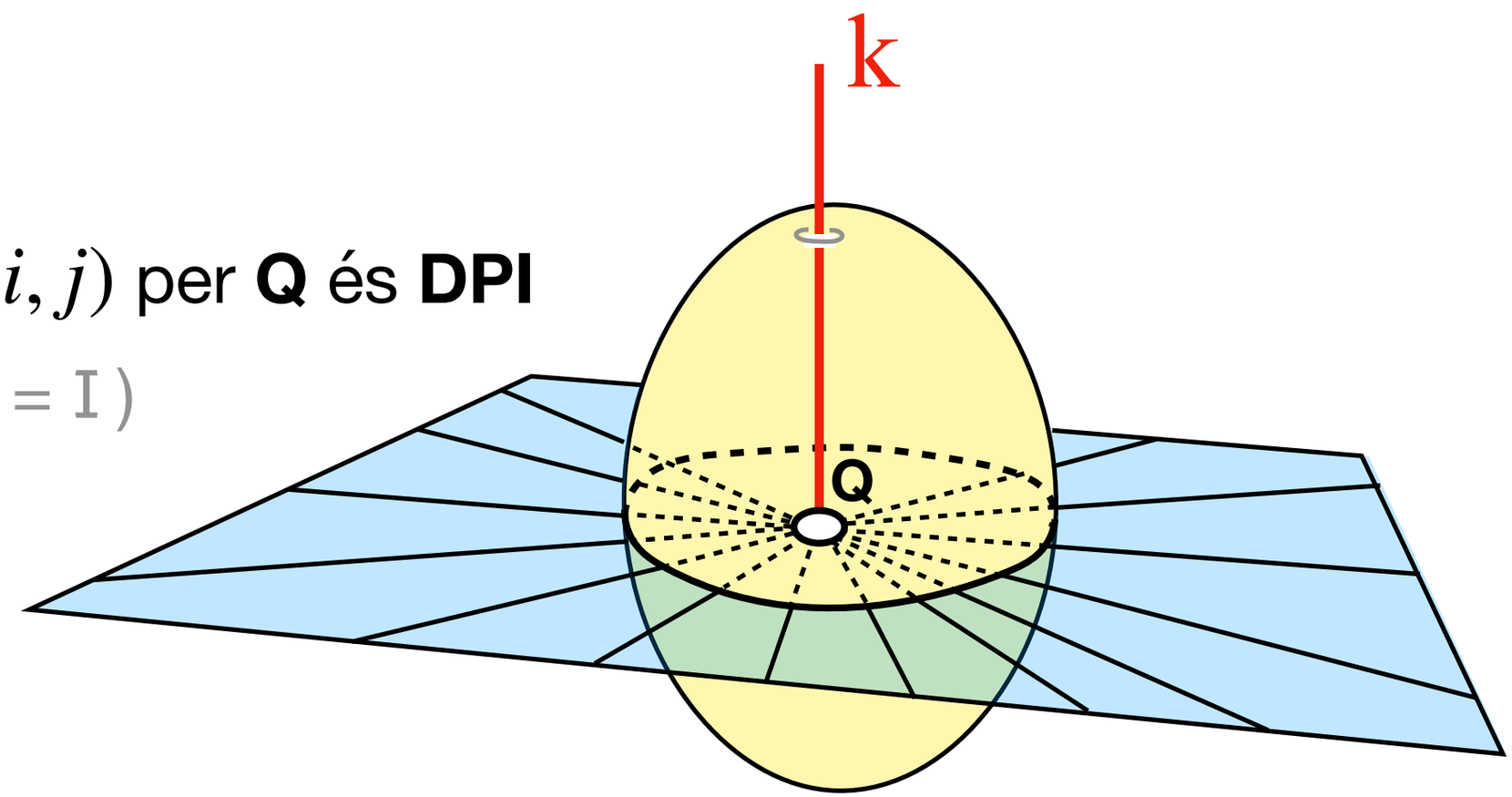
"Rotor simètric per **Q**" en el pla (i, j)

Si per al punt **Q** les dirs. (i, j)
són DPI amb mateix MPI ...

$$[\mathbb{I}(\mathbf{Q})]_B = \begin{bmatrix} \mathtt{I} & 0 & 0 \\ 0 & \mathtt{I} & 0 \\ 0 & 0 & \mathtt{I}_{kk} \end{bmatrix}$$

... tota recta del pla (i, j) per **Q** és **DPI**

(amb MPI = $\mathtt{I}$)



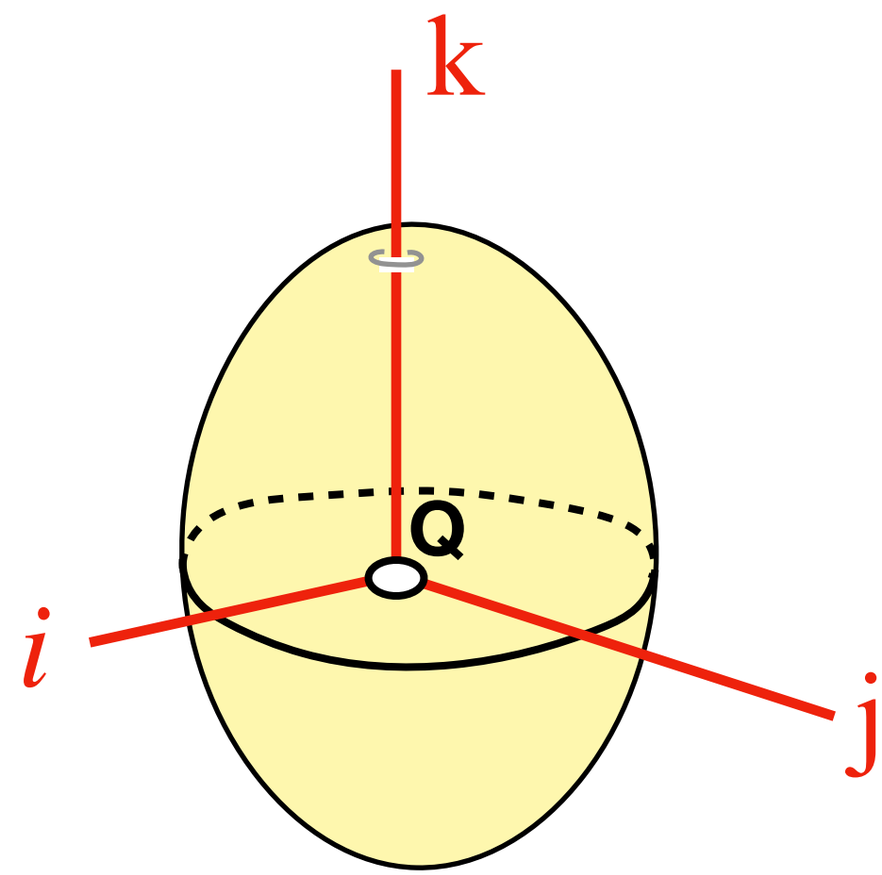
"Rotor esfèric per **Q**"

Si per al punt **Q** les dirs. (i, j, k)
són **DPI** amb mateix **MPI** ...

... el tensor a **Q** té la forma

$$\begin{bmatrix} \mathtt{I} & 0 & 0 \\ 0 & \mathtt{I} & 0 \\ 0 & 0 & \mathtt{I} \end{bmatrix}$$

independentment de la base triada



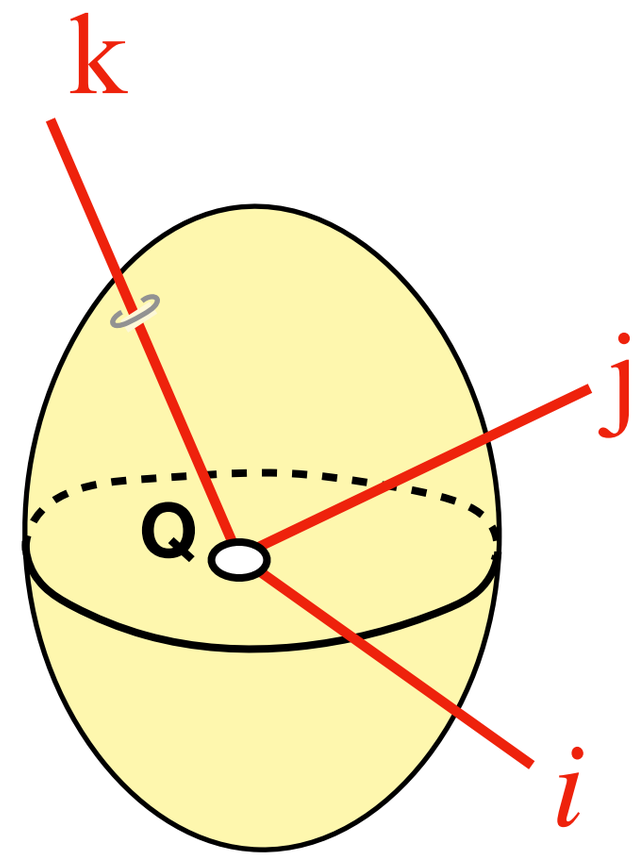
"Rotor esfèric per **Q**"

Si per al punt **Q** les dirs. (i, j, k)
són **DPI** amb mateix **MPI** ...

... el tensor a **Q** té la forma

$$\begin{bmatrix} \mathtt{I} & 0 & 0 \\ 0 & \mathtt{I} & 0 \\ 0 & 0 & \mathtt{I} \end{bmatrix}$$

independentment de la base triada



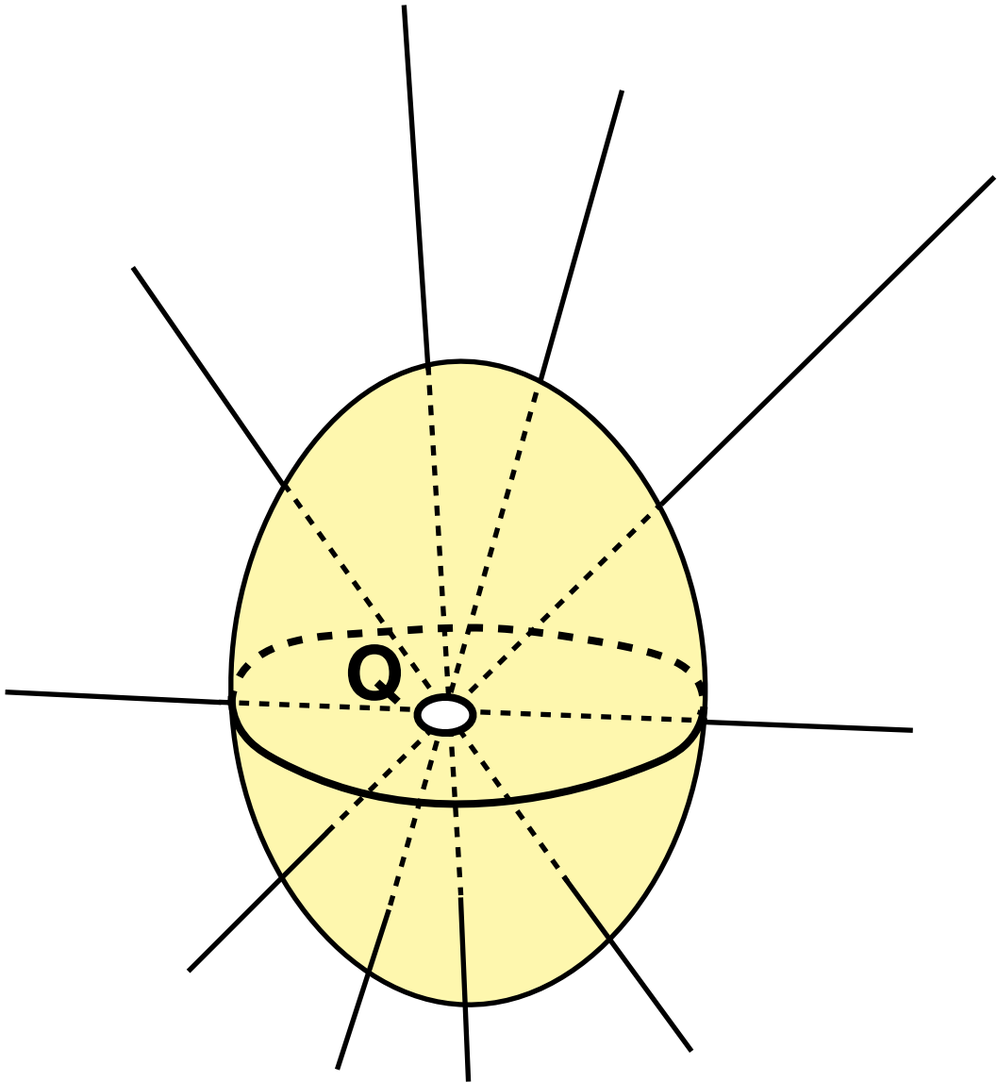
"Rotor esfèric per **Q**"

Si per al punt **Q** les dirs. (i, j, k) són **DPI** amb mateix **MPI** ...

... el tensor a **Q** té la forma

$$\begin{bmatrix} \mathtt{I} & 0 & 0 \\ 0 & \mathtt{I} & 0 \\ 0 & 0 & \mathtt{I} \end{bmatrix}$$

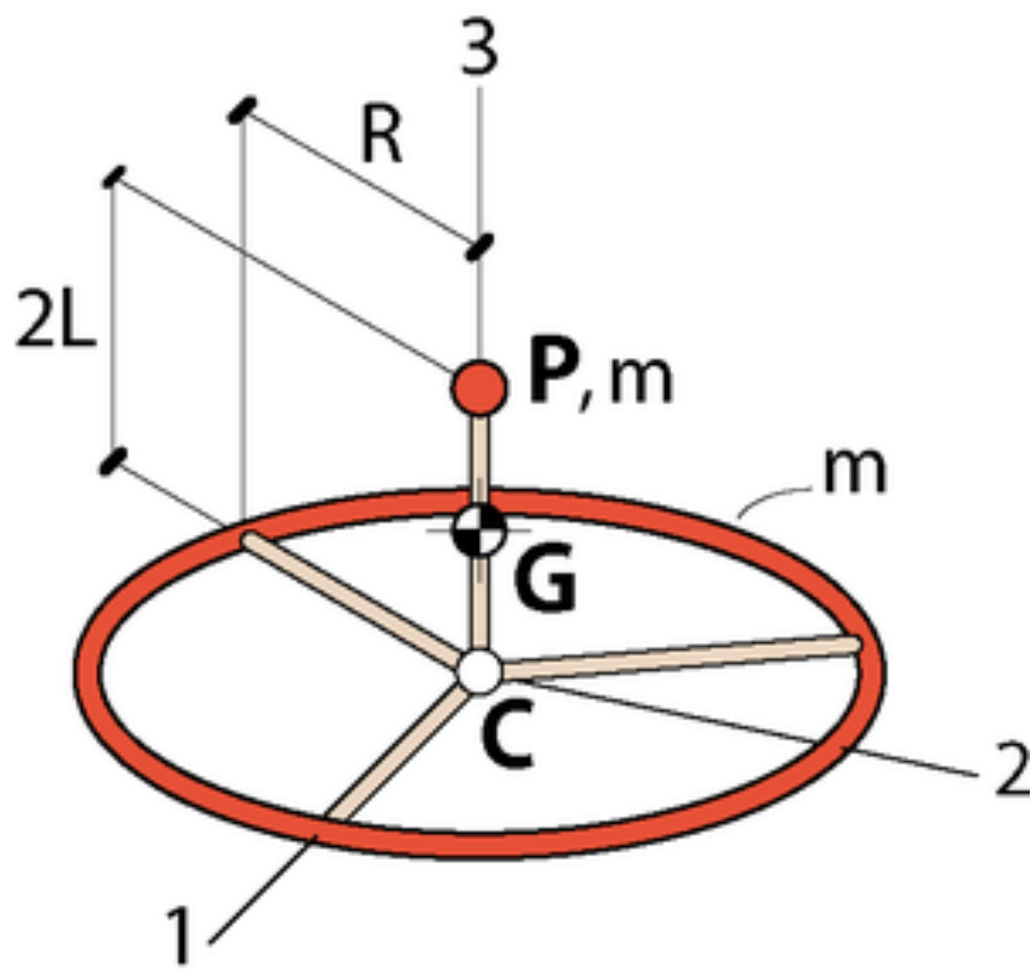
independentment de la base triada



Qualsevol recta per **Q** és **DPI**

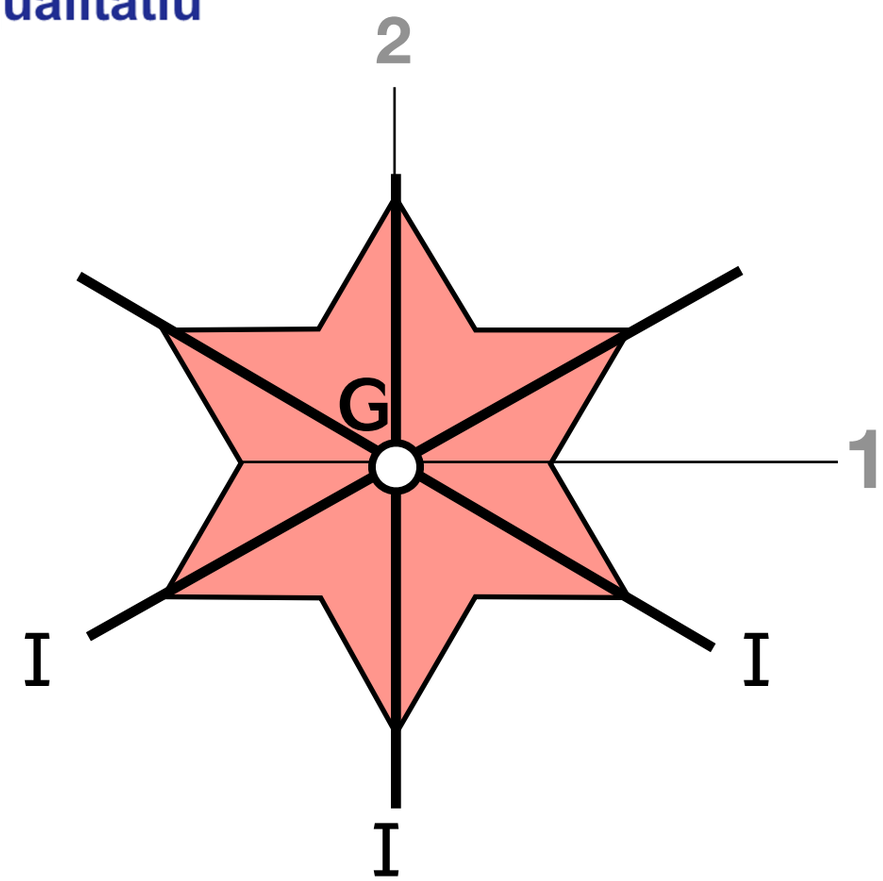
(amb MPI = $\mathtt{I}$)

Exemple D5.9 - Wikimec



$[\mathbb{I}(\mathbf{G})]$?

qualitatiu



Sòlid pla i eix 2 de simetria

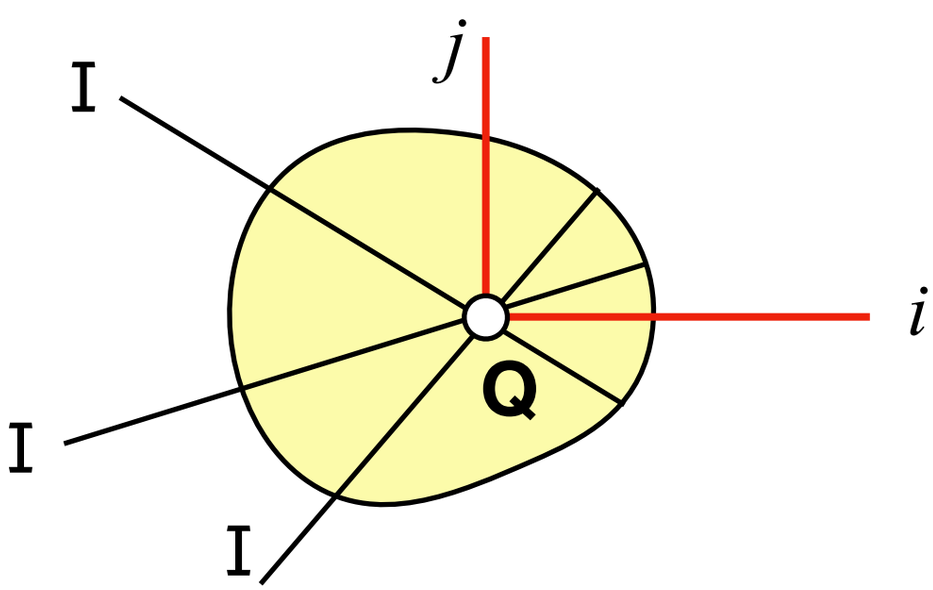
$$[\mathbb{I}(\mathbf{G})]_B = \begin{bmatrix} I_{11} & & \\ & I_{22} & \\ & & I_{11}+I_{22} \end{bmatrix}$$

I_{11}, I_{22} ? $\longleftarrow$ Són iguals!

3 moments en pla (1,2) iguals $\Rightarrow$ Rotor simètric a **G** per aquest pla!

$$[\mathbb{I}(\mathbf{G})]_B = \begin{bmatrix} I & & \\ & I & \\ & & 2I \end{bmatrix}$$

Si **3 o més** moments d'inèrcia resp. eixos d'un **mateix pla** (i,j) són iguals ...



... el sòlid és **rotor simètric a Q** per aquest pla

Avaluació quantitativa

$[\mathbb{I}(\mathbf{O})]$?

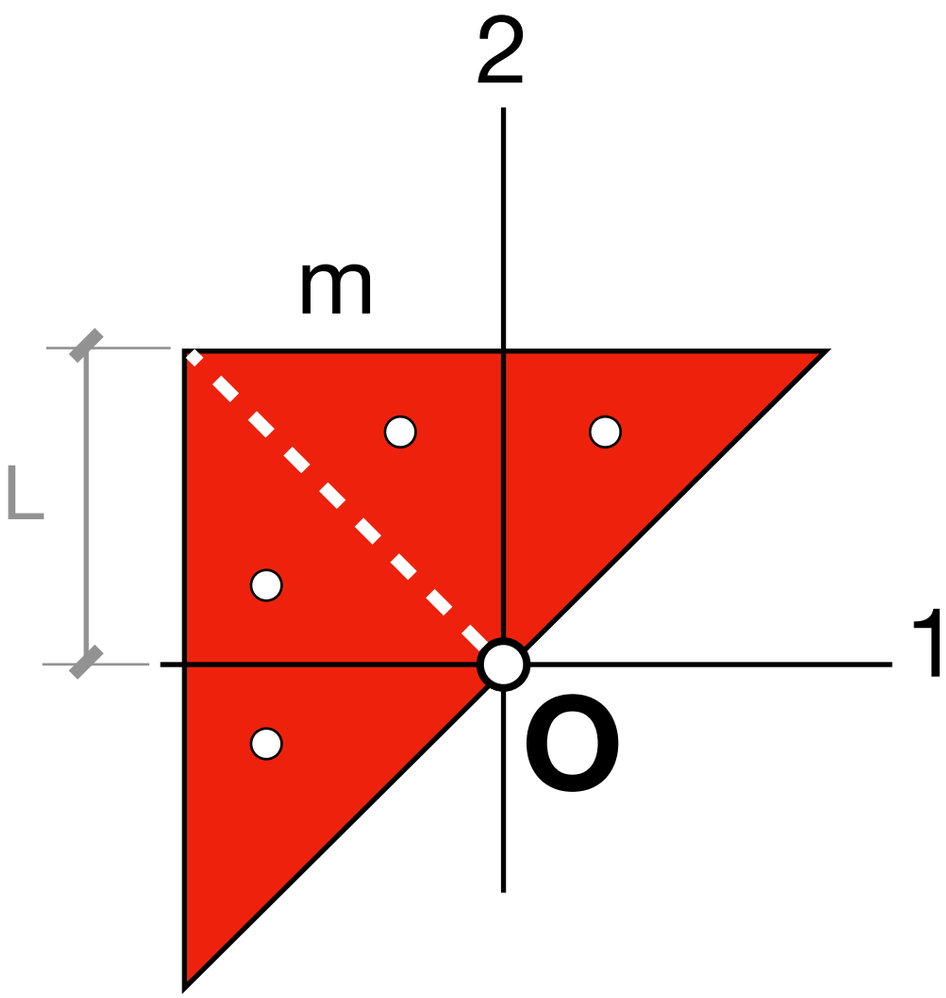
qualitatiu

quantitatiu

Fig. plana $\Longrightarrow$ 3 és DPI

$\mathtt{I}_{12}$? ← És zero!

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} \mathtt{I}_{11} & & \\ & \mathtt{I}_{22} & \\ & & \mathtt{I}_{11} + \mathtt{I}_{22} \end{bmatrix}$$



$[\mathbb{I}(\mathbf{O})]$?

qualitatiu

quantitatiu

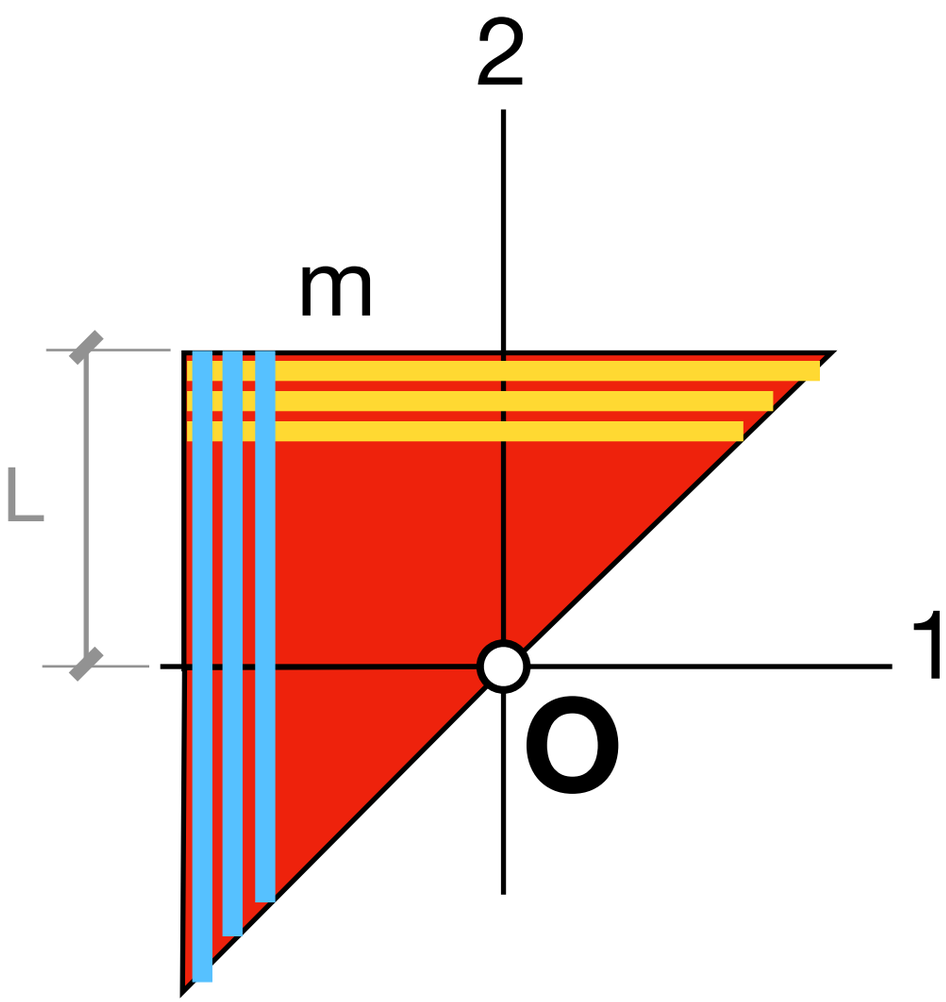


Fig. plana $\Longrightarrow$ 3 és DPI

I_{12}? ← És zero!

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} \mathrm{I}_{11} & & \\ & \mathrm{I}_{22} & \\ & & \mathrm{I}_{11} + \mathrm{I}_{22} \end{bmatrix}$$

$\mathrm{I}_{11}, \mathrm{I}_{22}$? ← Són iguals!

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} \mathrm{I} & & \\ & \mathrm{I} & \\ & & 2\mathrm{I} \end{bmatrix}$$

Rotor simètric a **O**

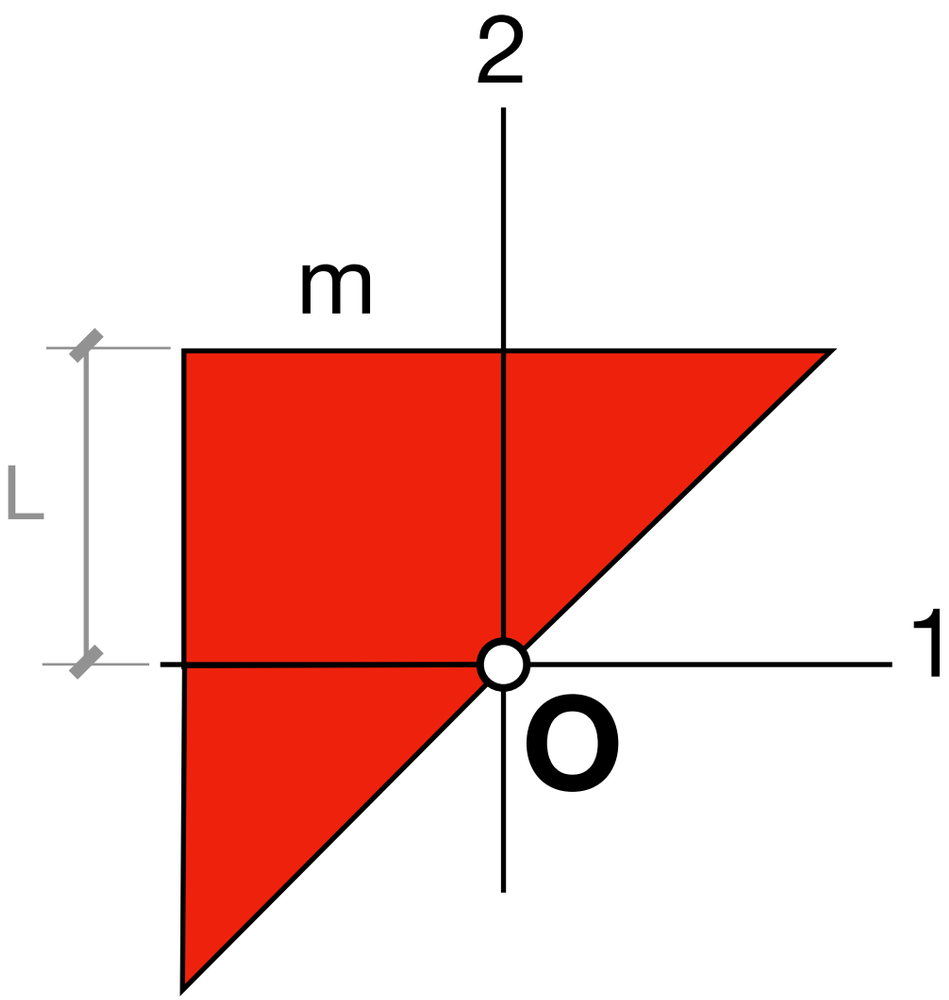
$[\mathbb{I}(\mathbf{O})]$?

qualitatiu

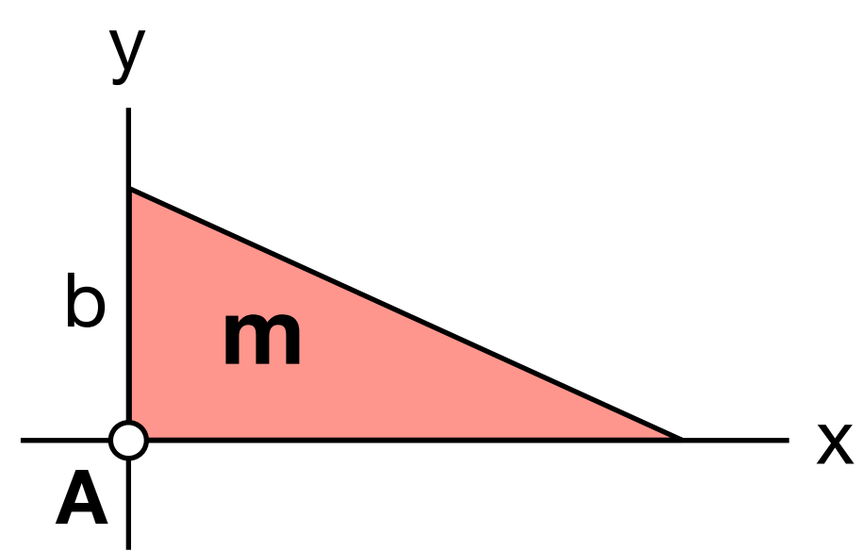
quantitatiu

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} \mathtt{I} & & \\ & \mathtt{I} & \\ & & 2\mathtt{I} \end{bmatrix}$$

Quant val $\mathtt{I}$?



Taula



$$\mathtt{I}_{xx}(\mathbf{A}) = \frac{1}{6} m b^2$$

Taula no aplicable directament

(**O** no és el vèrtex de 90º)

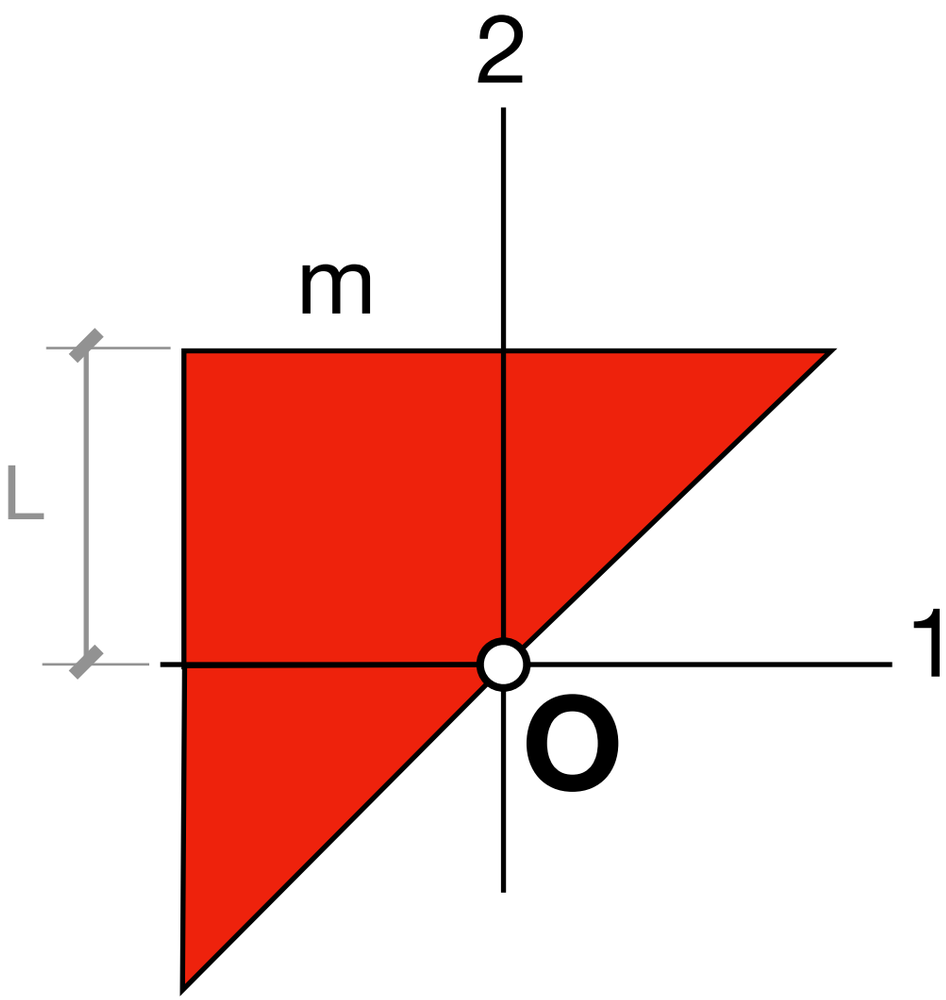
$[\mathbb{I}(\mathbf{O})]$?

qualitatiu

quantitatiu

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} I & & \\ & I & \\ & & 2I \end{bmatrix}$$

Quant val I ?



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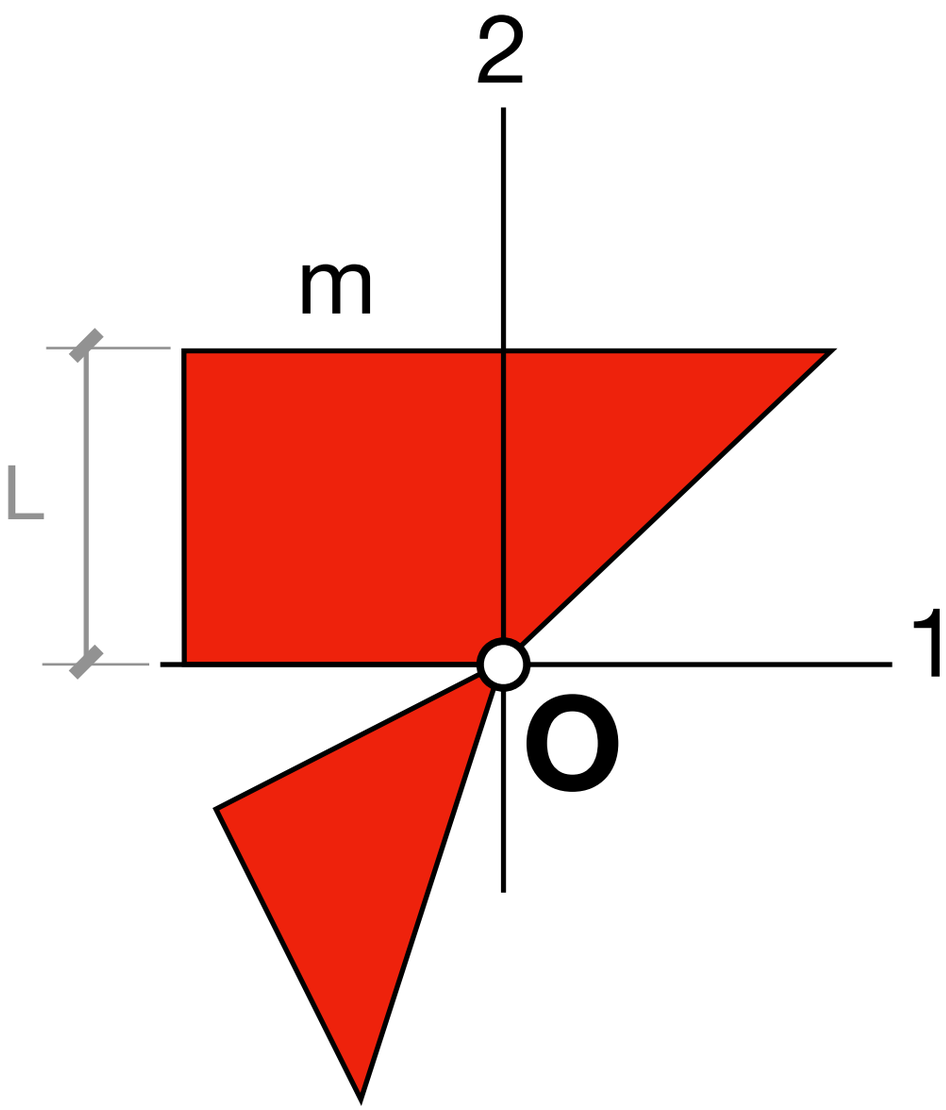

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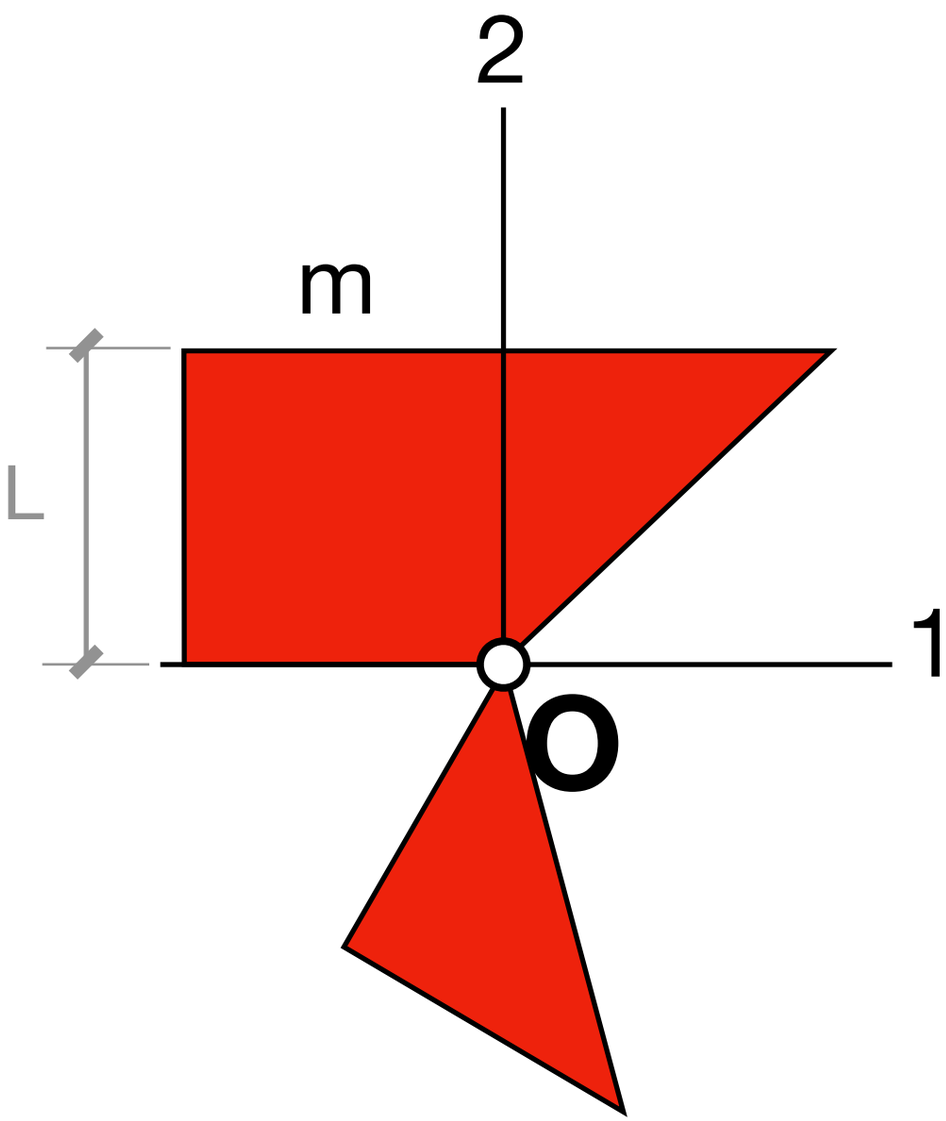

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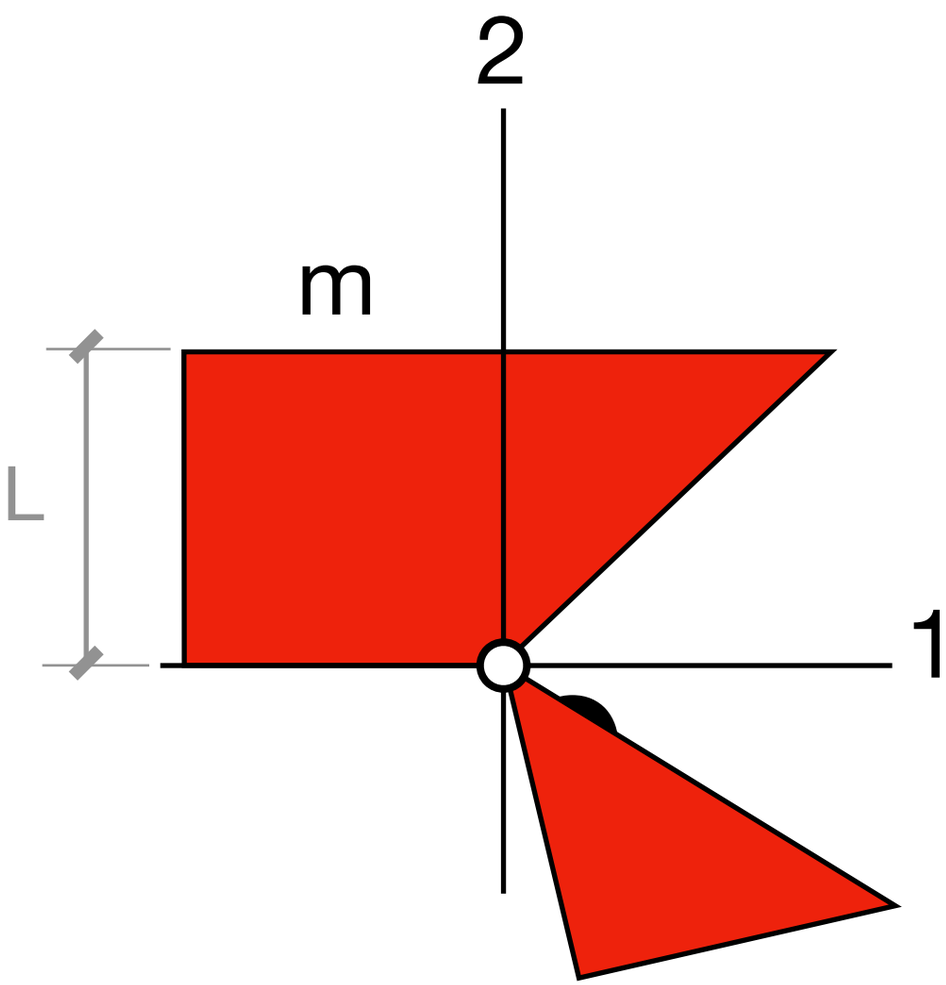

$[\mathbb{I}(\mathbf{O})]$?

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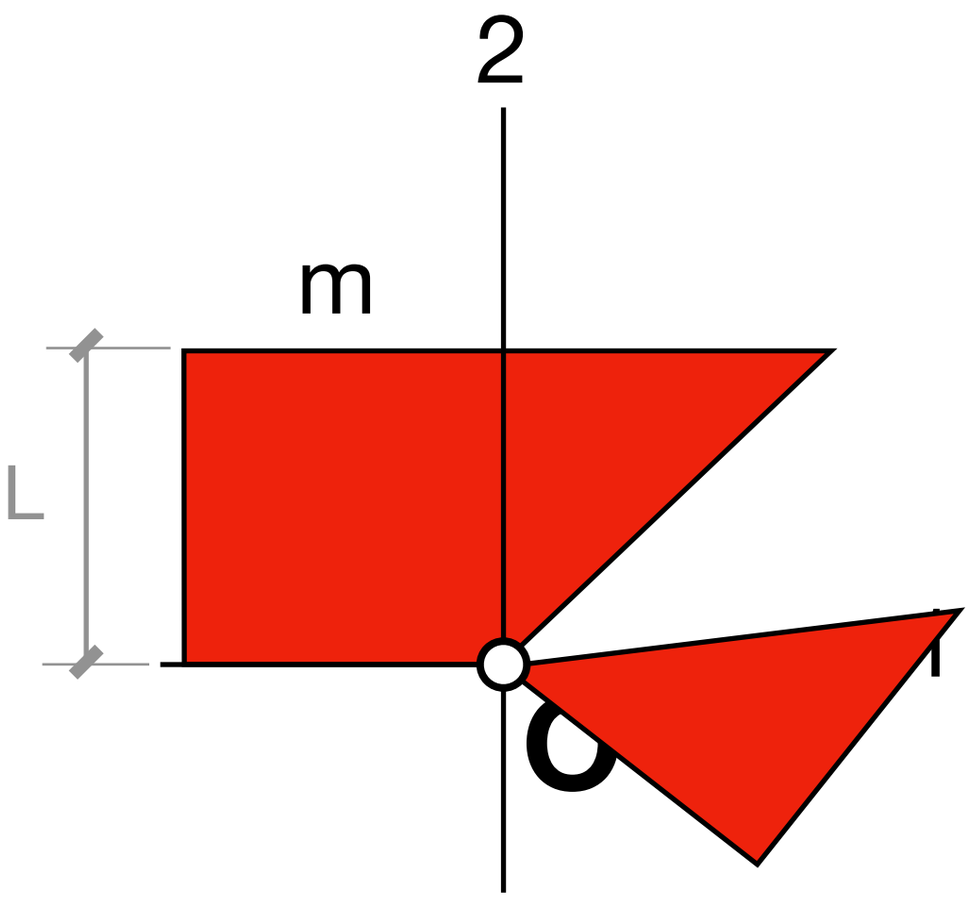
Quant val I ?



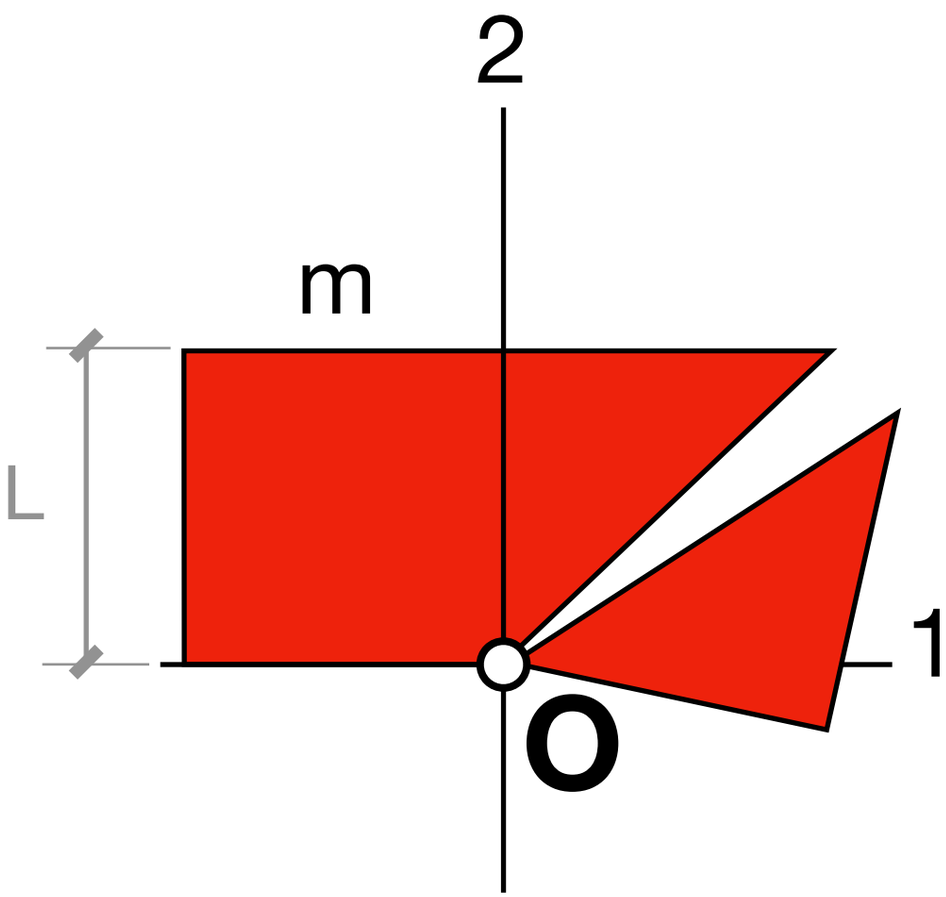
$[\mathbb{I}(\mathbf{O})]$?

qualitatiu

quantitatiu

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} \mathtt{I} & & \\ & \mathtt{I} & \\ & & 2\mathtt{I} \end{bmatrix}$$

Quant val $\mathtt{I}$?



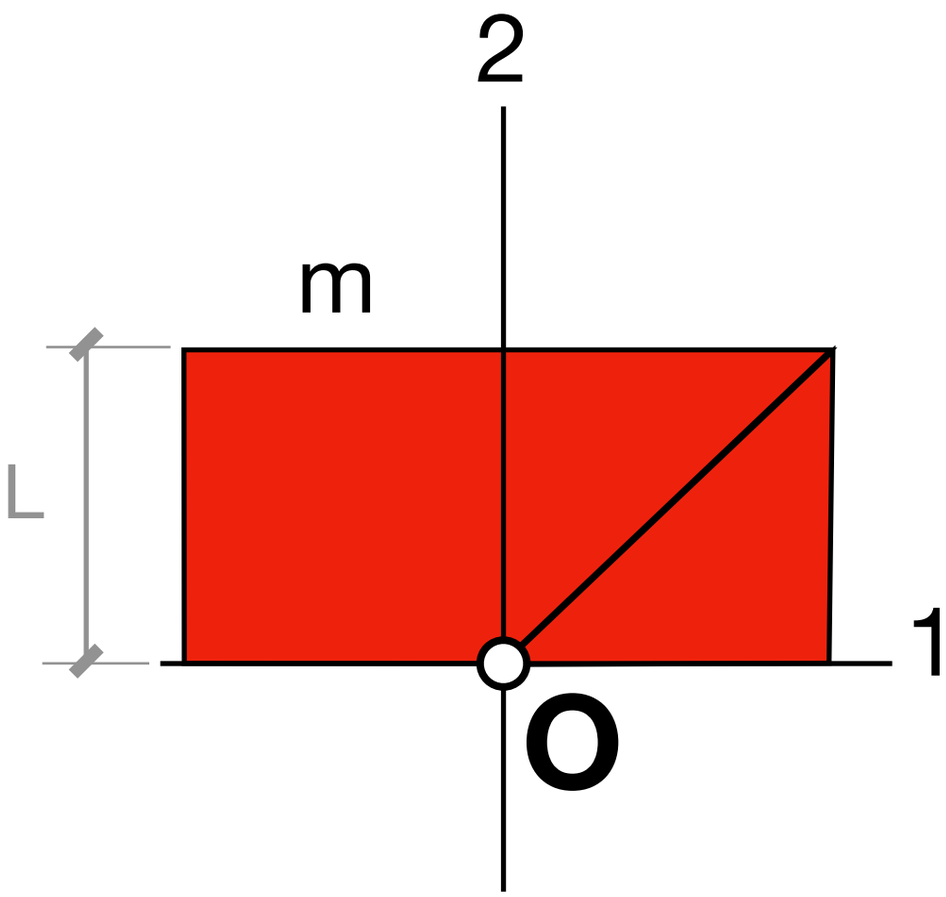
$[\mathbb{I}(\mathbf{O})]$?

qualitatiu

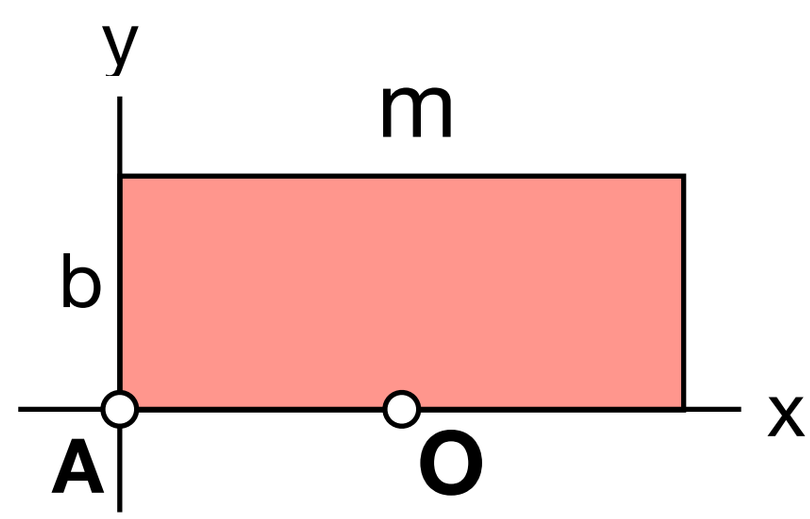
quantitatiu

$$[\mathbb{I}(\mathbf{O})]_B = \begin{bmatrix} I & & \\ & I & \\ & & 2I \end{bmatrix}$$

Quant val I ?



Taula



$$I_{xx}(\mathbf{A}) = \frac{1}{3} m b^2$$

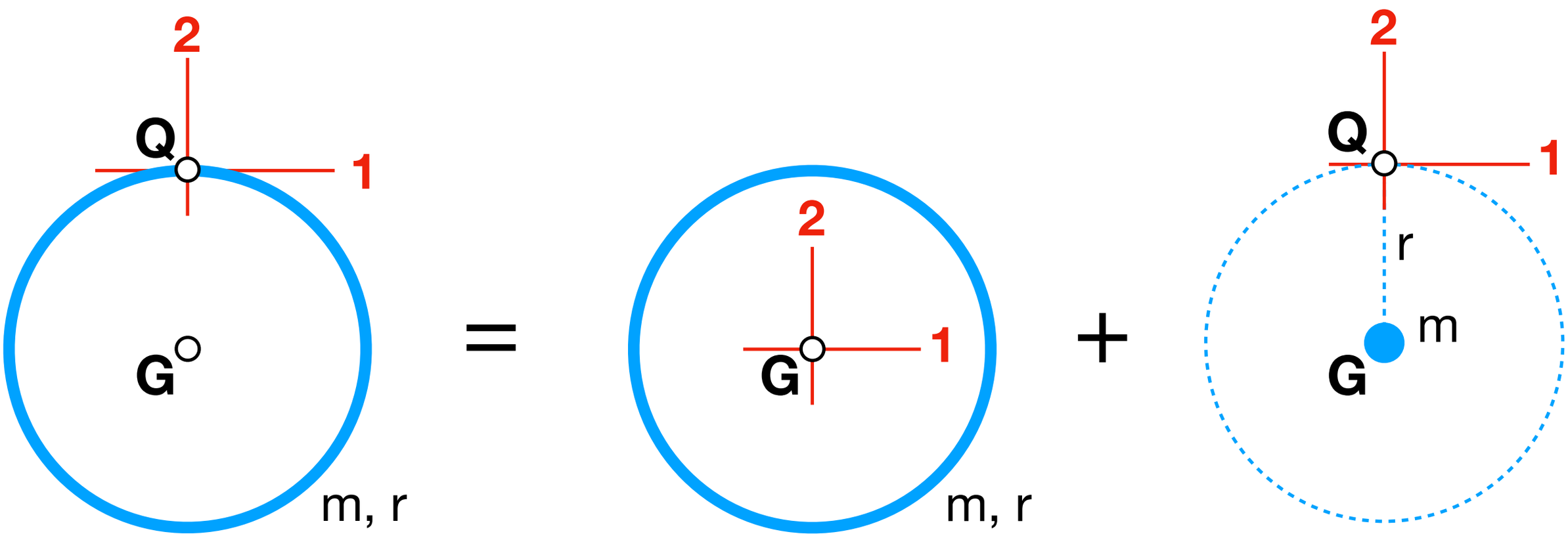
$$I = \frac{1}{3} m L^2$$

Teorema de Steiner

$$\underbrace{\mathbb{I}(\mathbf{Q})}_{\text{Tensor a } \mathbf{Q}} = \underbrace{\mathbb{I}(\mathbf{G})}_{\text{Tensor a } \mathbf{G}} + \underbrace{\mathbb{I}^{\oplus}(\mathbf{Q})}_{\text{Tensor a } \mathbf{Q} \text{ de tota la massa concentrada a } \mathbf{G}}$$

Exm: anell homogeni

$[\mathbb{I}(\mathbf{Q})]_B$?



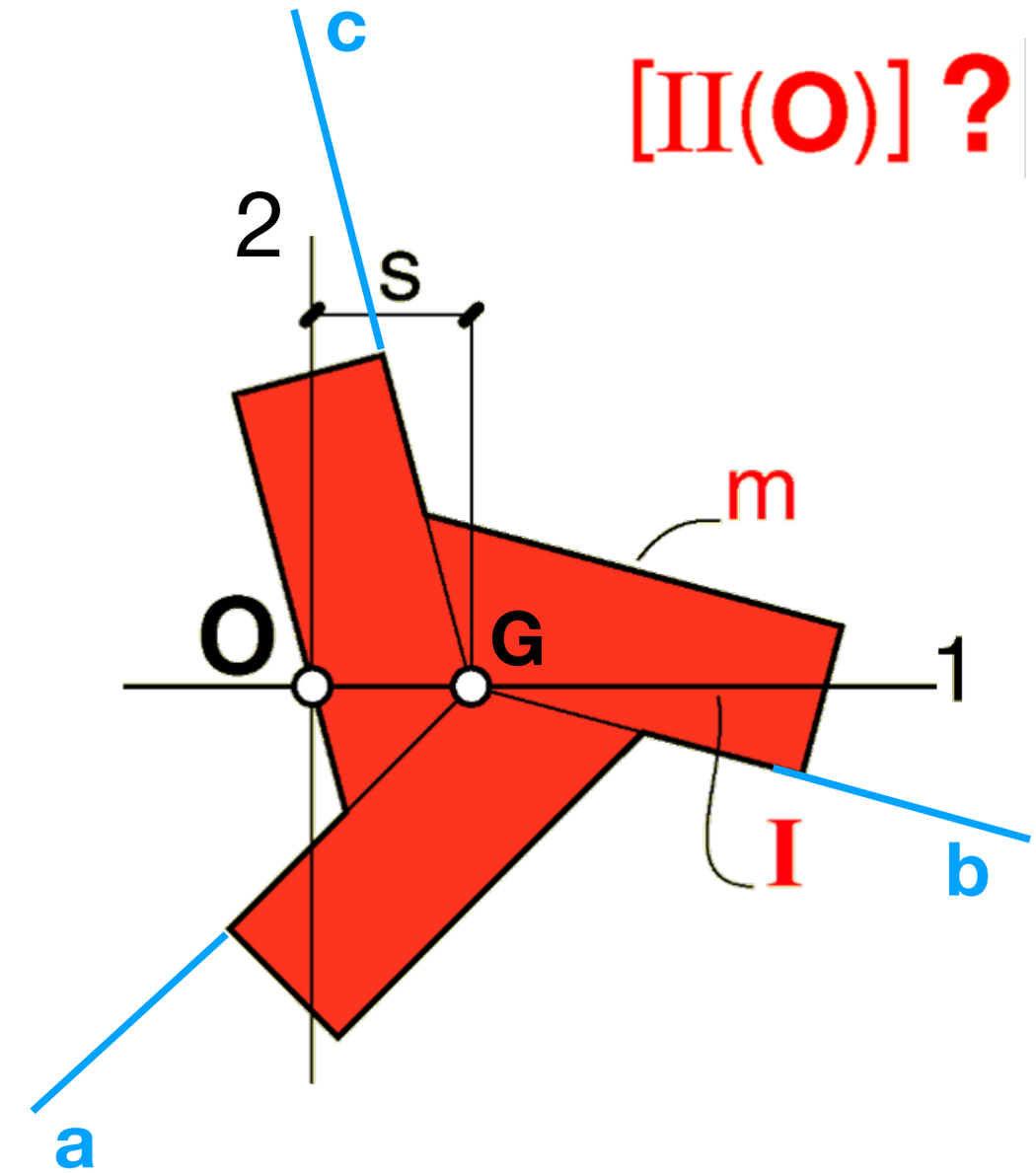
Tensor a G és fàcil:

Sòlid pla $\Longrightarrow$ 3 és DPI

$$\underbrace{I_{aa} = I_{bb} = I_{cc}} = I$$

rotor simètric per a G

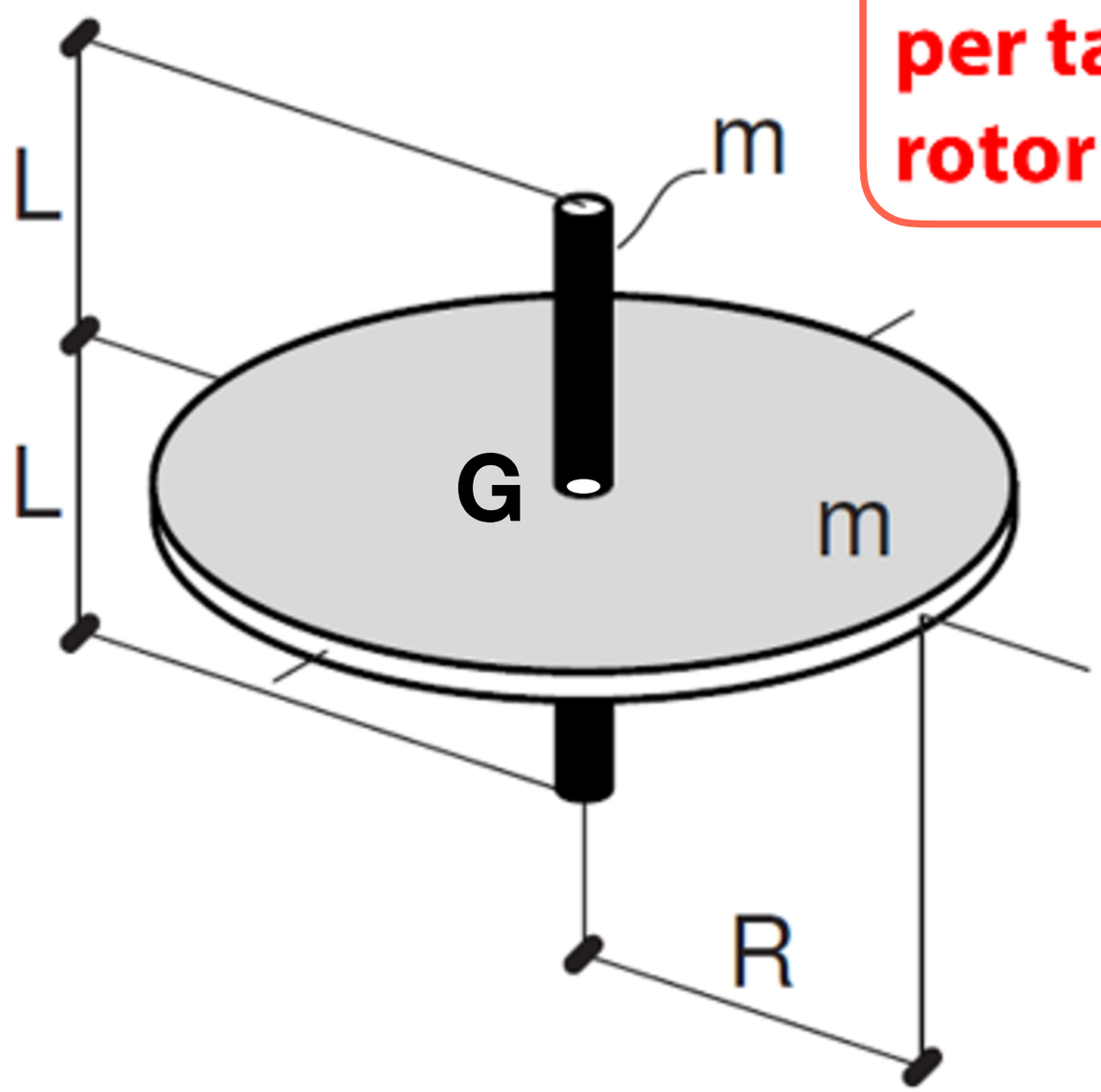
$$[\mathbb{I}(G)]_B = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 2I \end{bmatrix}$$

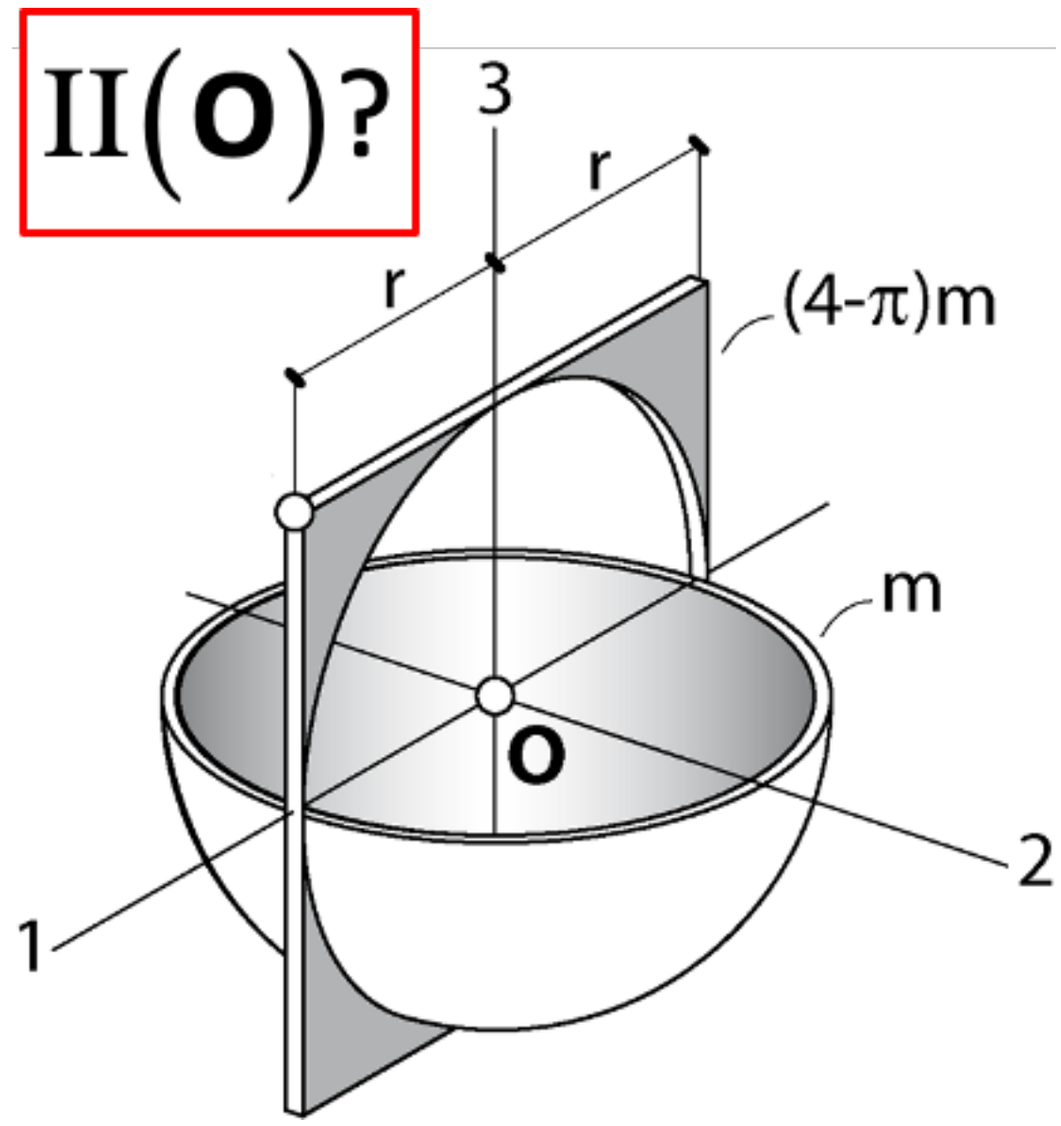


Canvi a O via Steiner:

$$[\mathbb{I}(O)]_B = \underbrace{\begin{bmatrix} I & & \\ & I & \\ & & 2I \end{bmatrix}}_{\mathbb{I}(G)} + \underbrace{\begin{bmatrix} 0 & & \\ & ms^2 & \\ & & ms^2 \end{bmatrix}}_{\mathbb{I}^{\oplus}(O)}$$

Deures:



Relació entre L i R per tal que sigui rotor esfèric a G?

Deures: